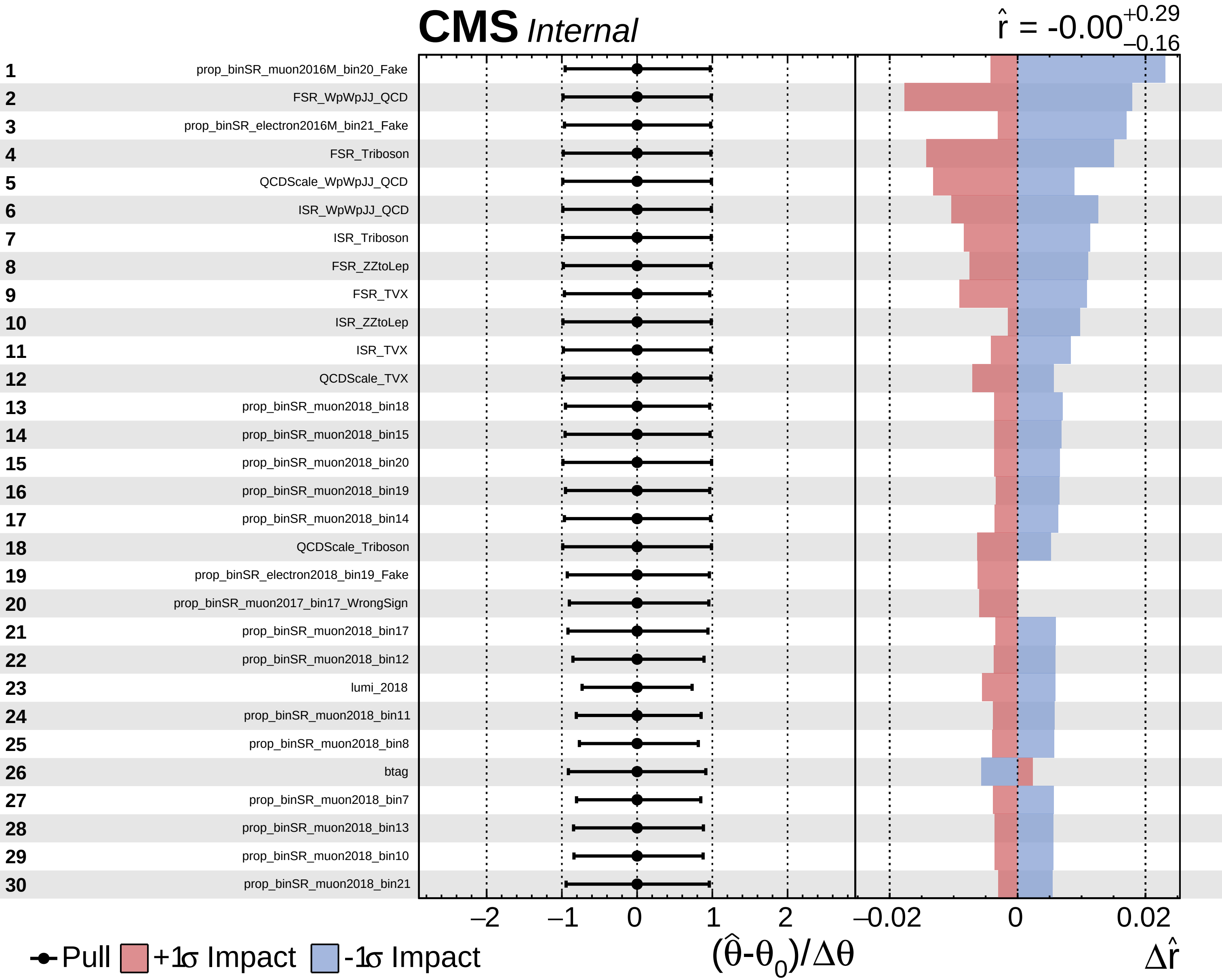
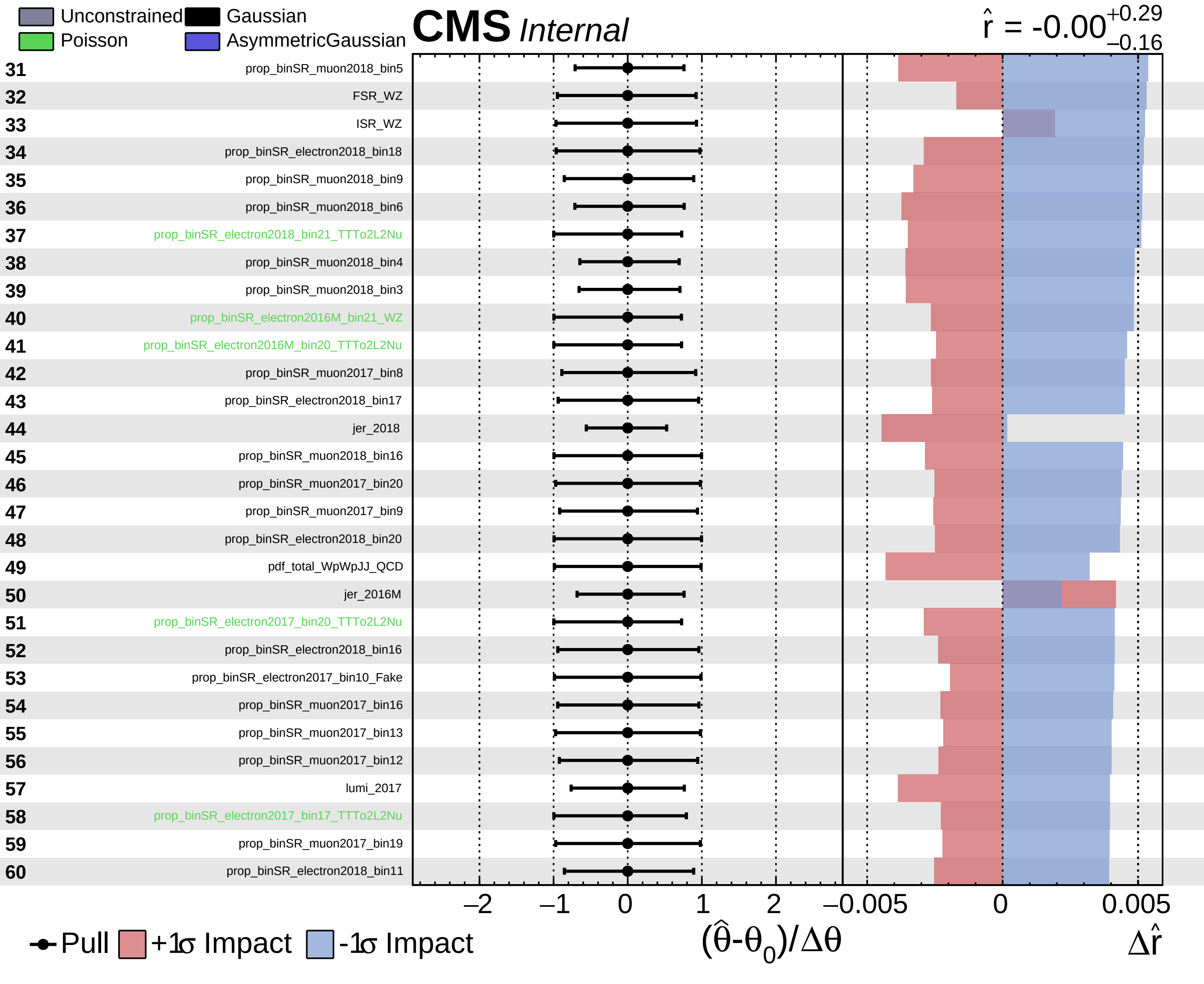
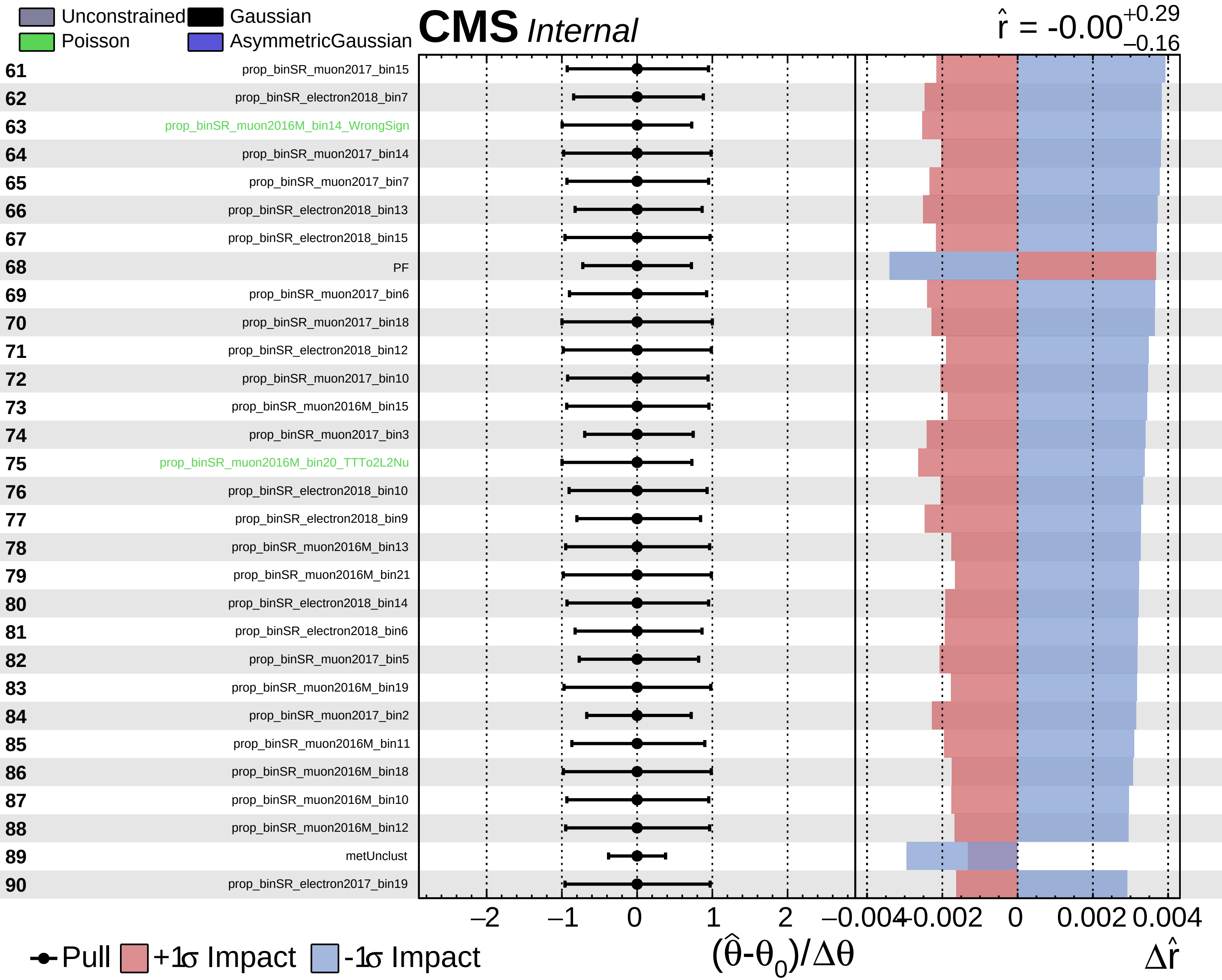


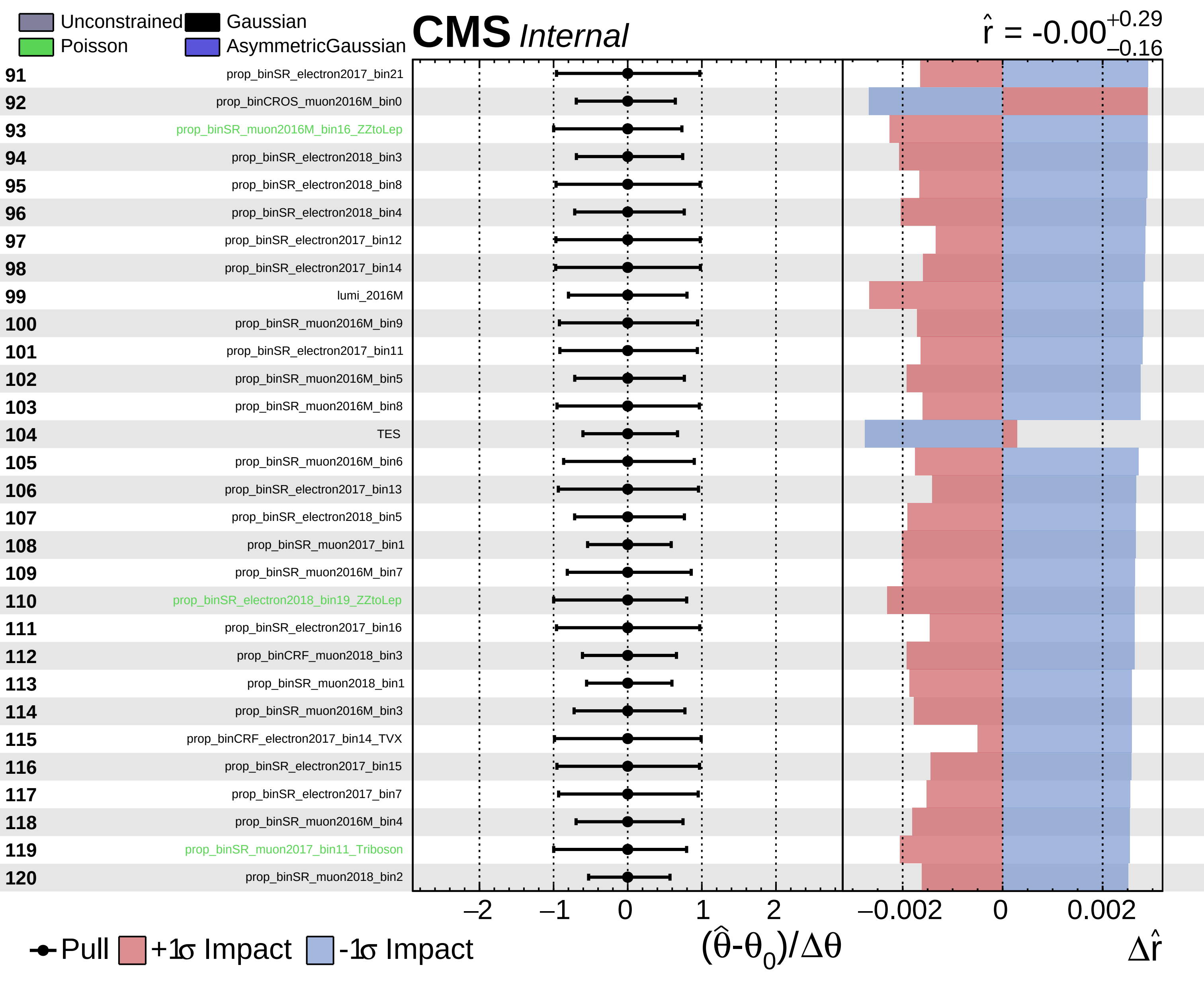
CMS Internal
$\hat{r} = -0.00^{+0.29}_{-0.16}$
1 prop_binSR_muon2016M_bin20_Fake
2 FSR_WpWpJJ_QCD
3 prop_binSR_electron2016M_bin21_Fake
4 FSR_Triboson
5 QCDScale_WpWpJJ_QCD
6 ISR_WpWpJJ_QCD
7 ISR_Triboson
8 FSR_ZZtoLep
9 FSR_TVX
10 ISR_ZZtoLep
11 ISR_TVX
12 QCDScale_TVX
13 prop_binSR_muon2018_bin18
14 prop_binSR_muon2018_bin15
15 prop_binSR_muon2018_bin20
16 prop_binSR_muon2018_bin19
17 prop_binSR_muon2018_bin14
18 QCDScale_Triboson
19 prop_binSR_electron2018_bin19_Fake
20 prop_binSR_muon2017_bin17_WrongSign
21 prop_binSR_muon2018_bin17
22 prop_binSR_muon2018_bin12
23 lumi_2018
24 prop_binSR_muon2018_bin11
25 prop_binSR_muon2018_bin8
26 btag
27 prop_binSR_muon2018_bin7
28 prop_binSR_muon2018_bin13
29 prop_binSR_muon2018_bin10
30 prop_binSR_muon2018_bin21
−2
−1
0
1
2
−0.02
0
0.02
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



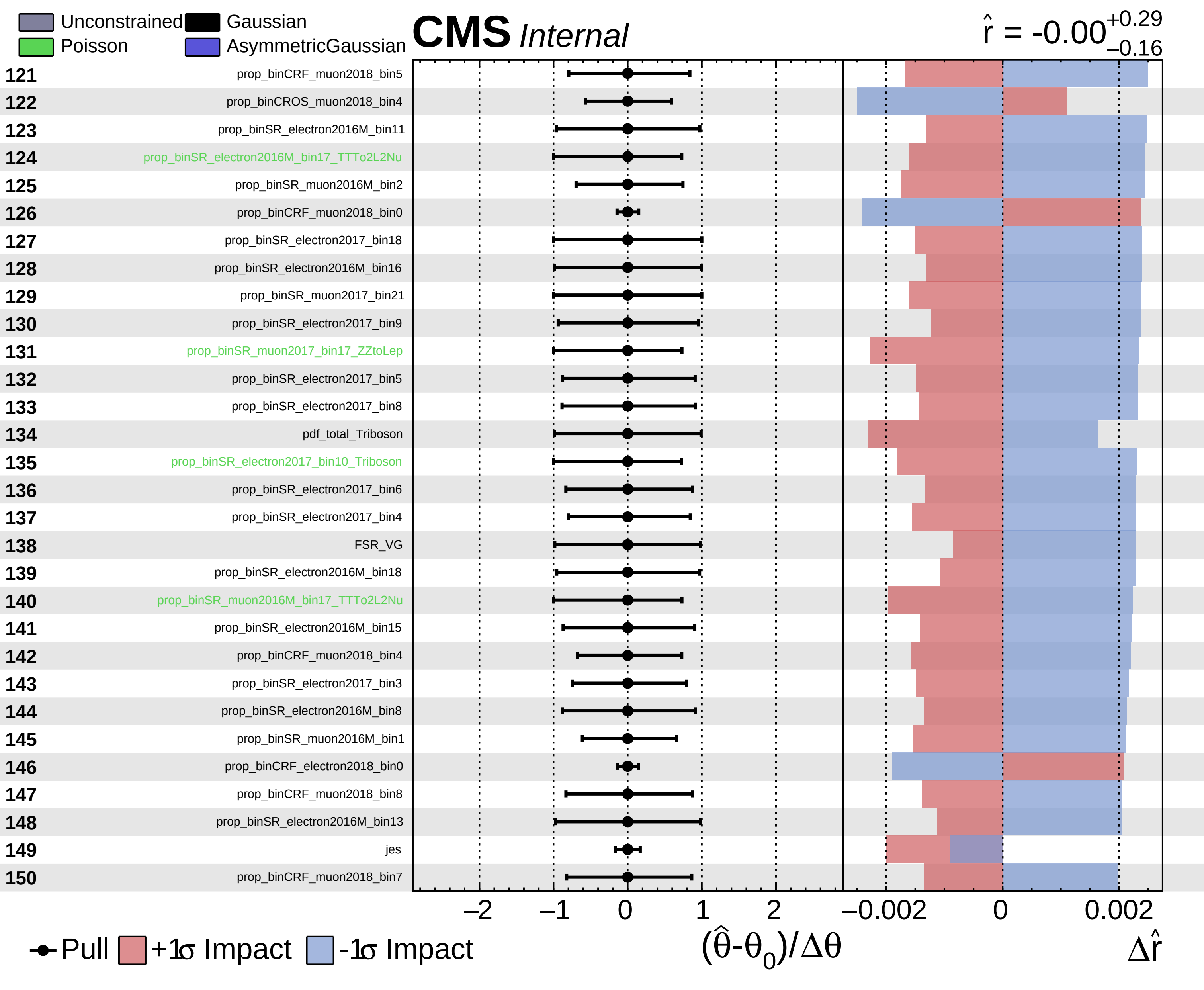
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.29}_{-0.16}$
31 prop_binSR_muon2018_bin5
32 FSR_WZ
33 ISR_WZ
34 prop_binSR_electron2018_bin18
35 prop_binSR_muon2018_bin9
36 prop_binSR_muon2018_bin6
37 prop_binSR_electron2018_bin21_TTTo2L2Nu
38 prop_binSR_muon2018_bin4
39 prop_binSR_muon2018_bin3
40 prop_binSR_electron2016M_bin21_WZ
41 prop_binSR_electron2016M_bin20_TTTo2L2Nu
42 prop_binSR_muon2017_bin8
43 prop_binSR_electron2018_bin17
44 jer_2018
45 prop_binSR_muon2018_bin16
46 prop_binSR_muon2017_bin20
47 prop_binSR_muon2017_bin9
48 prop_binSR_electron2018_bin20
49 pdf_total_WpWpJJ_QCD
50 jer_2016M
51 prop_binSR_electron2017_bin20_TTTo2L2Nu
52 prop_binSR_electron2018_bin16
53 prop_binSR_electron2017_bin10_Fake
54 prop_binSR_muon2017_bin16
55 prop_binSR_muon2017_bin13
56 prop_binSR_muon2017_bin12
57 lumi_2017
58 prop_binSR_electron2017_bin17_TTTo2L2Nu
59 prop_binSR_muon2017_bin19
60 prop_binSR_electron2018_bin11
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



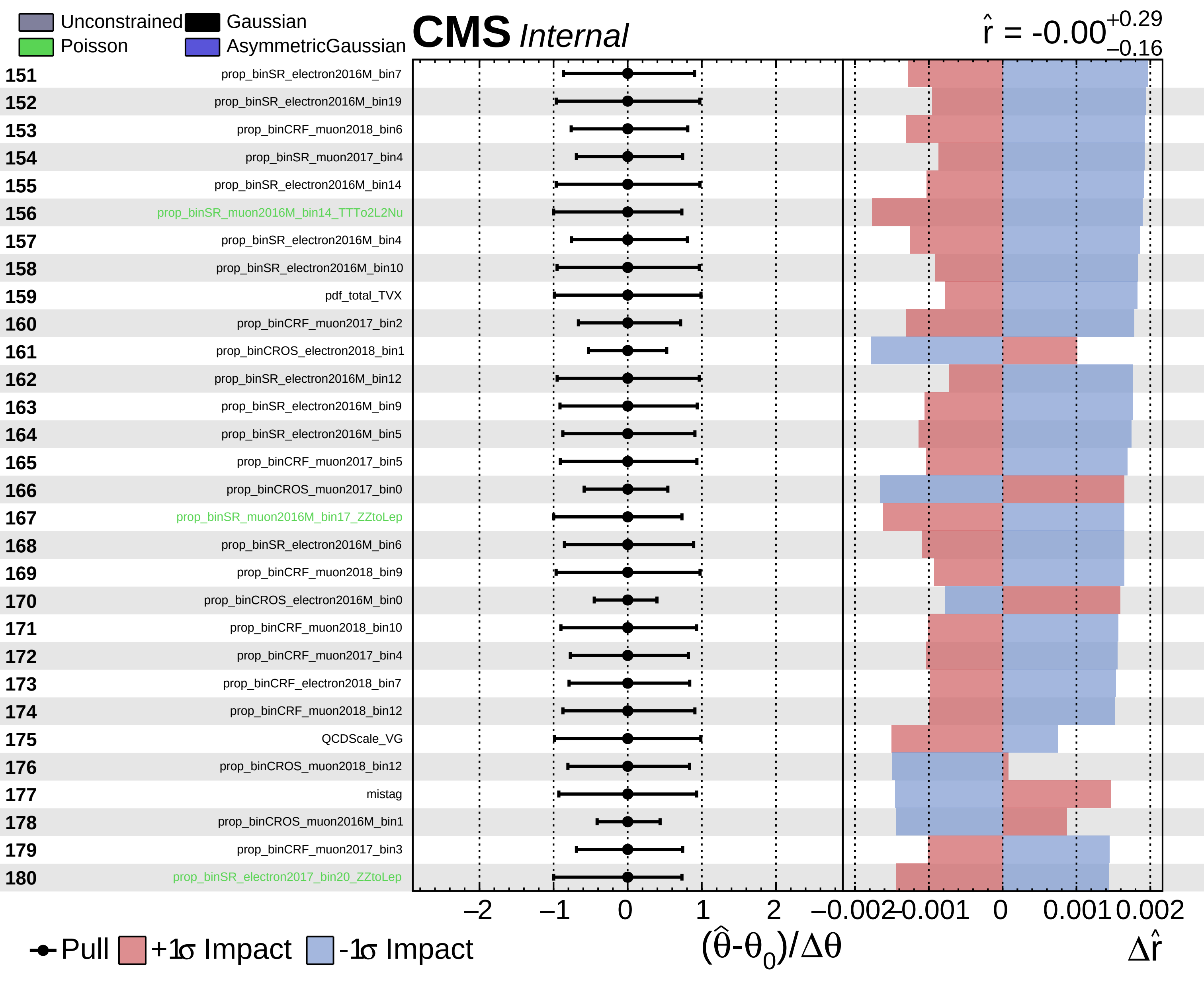
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.29}_{-0.16}$
61 prop_binSR_muon2017_bin15
62 prop_binSR_electron2018_bin7
63 prop_binSR_muon2016M_bin14_WrongSign
64 prop_binSR_muon2017_bin14
65 prop_binSR_muon2017_bin7
66 prop_binSR_electron2018_bin13
67 prop_binSR_electron2018_bin15
68 PF
69 prop_binSR_muon2017_bin6
70 prop_binSR_muon2017_bin18
71 prop_binSR_electron2018_bin12
72 prop_binSR_muon2017_bin10
73 prop_binSR_muon2016M_bin15
74 prop_binSR_muon2017_bin3
75 prop_binSR_muon2016M_bin20_TTTo2L2Nu
76 prop_binSR_electron2018_bin10
77 prop_binSR_electron2018_bin9
78 prop_binSR_muon2016M_bin13
79 prop_binSR_muon2016M_bin21
80 prop_binSR_electron2018_bin14
81 prop_binSR_electron2018_bin6
82 prop_binSR_muon2017_bin5
83 prop_binSR_muon2016M_bin19
84 prop_binSR_muon2017_bin2
85 prop_binSR_muon2016M_bin11
86 prop_binSR_muon2016M_bin18
87 prop_binSR_muon2016M_bin10
88 prop_binSR_muon2016M_bin12
89 metUnclust
90 prop_binSR_electron2017_bin19
−2 −1 0 1 2
−0.004 −0.002 0 0.002 0.004
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



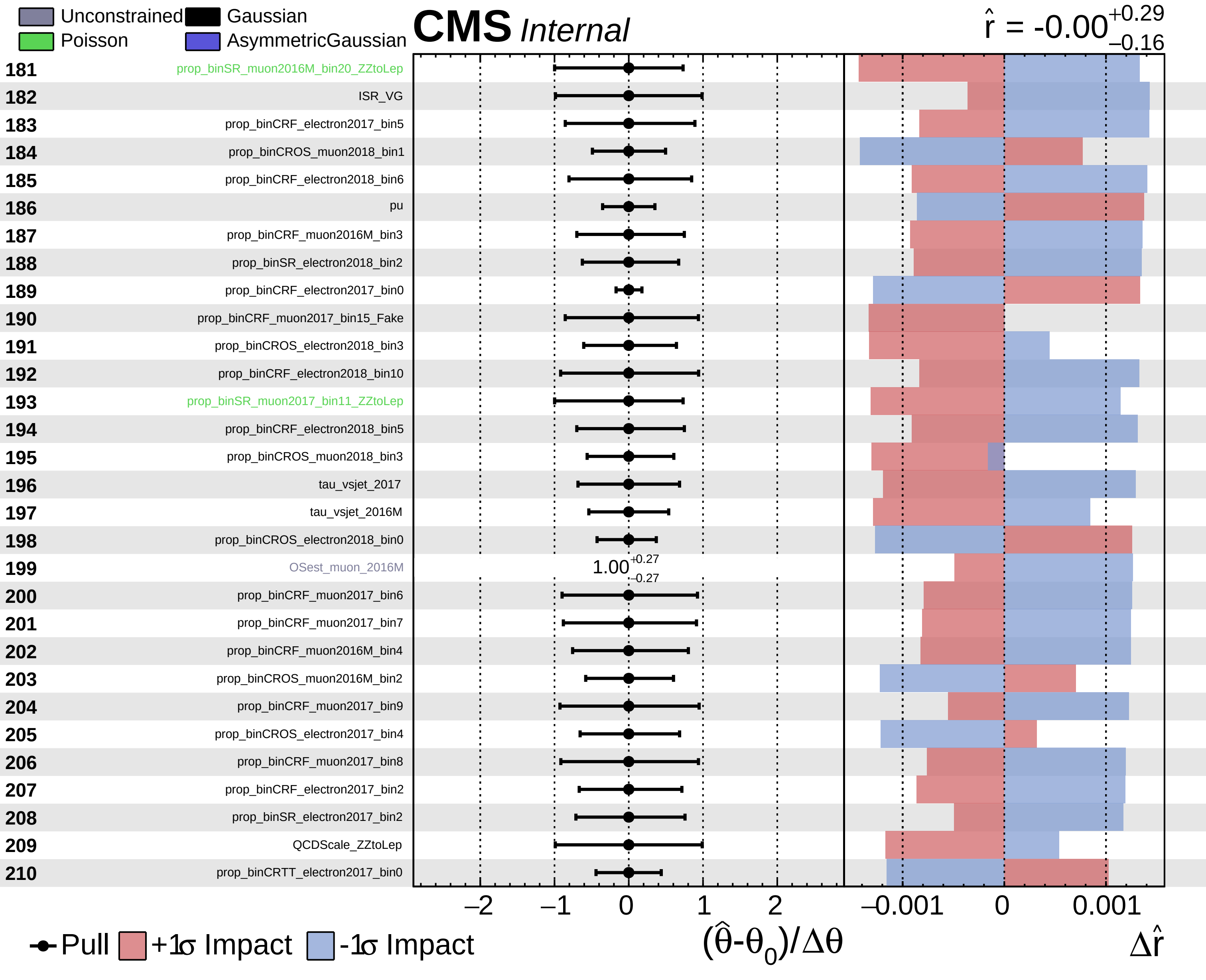
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.29}_{-0.16}$
91 prop_binSR_electron2017_bin21
92 prop_binCROS_muon2016M_bin0
93 prop_binSR_muon2016M_bin16_ZZtoLep
94 prop_binSR_electron2018_bin3
95 prop_binSR_electron2018_bin8
96 prop_binSR_electron2018_bin4
97 prop_binSR_electron2017_bin12
98 prop_binSR_electron2017_bin14
99 lumi_2016M
100 prop_binSR_muon2016M_bin9
101 prop_binSR_electron2017_bin11
102 prop_binSR_muon2016M_bin5
103 prop_binSR_muon2016M_bin8
104 TES
105 prop_binSR_muon2016M_bin6
106 prop_binSR_electron2017_bin13
107 prop_binSR_electron2018_bin5
108 prop_binSR_muon2017_bin1
109 prop_binSR_muon2016M_bin7
110 prop_binSR_electron2018_bin19_ZZtoLep
111 prop_binSR_electron2017_bin16
112 prop_binCRF_muon2018_bin3
113 prop_binSR_muon2018_bin1
114 prop_binSR_muon2016M_bin3
115 prop_binCRF_electron2017_bin14_TVX
116 prop_binSR_electron2017_bin15
117 prop_binSR_electron2017_bin7
118 prop_binSR_muon2016M_bin4
119 prop_binSR_muon2017_bin11_Triboson
120 prop_binSR_muon2018_bin2
−2 −1 0 1 2
−0.002 0 0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



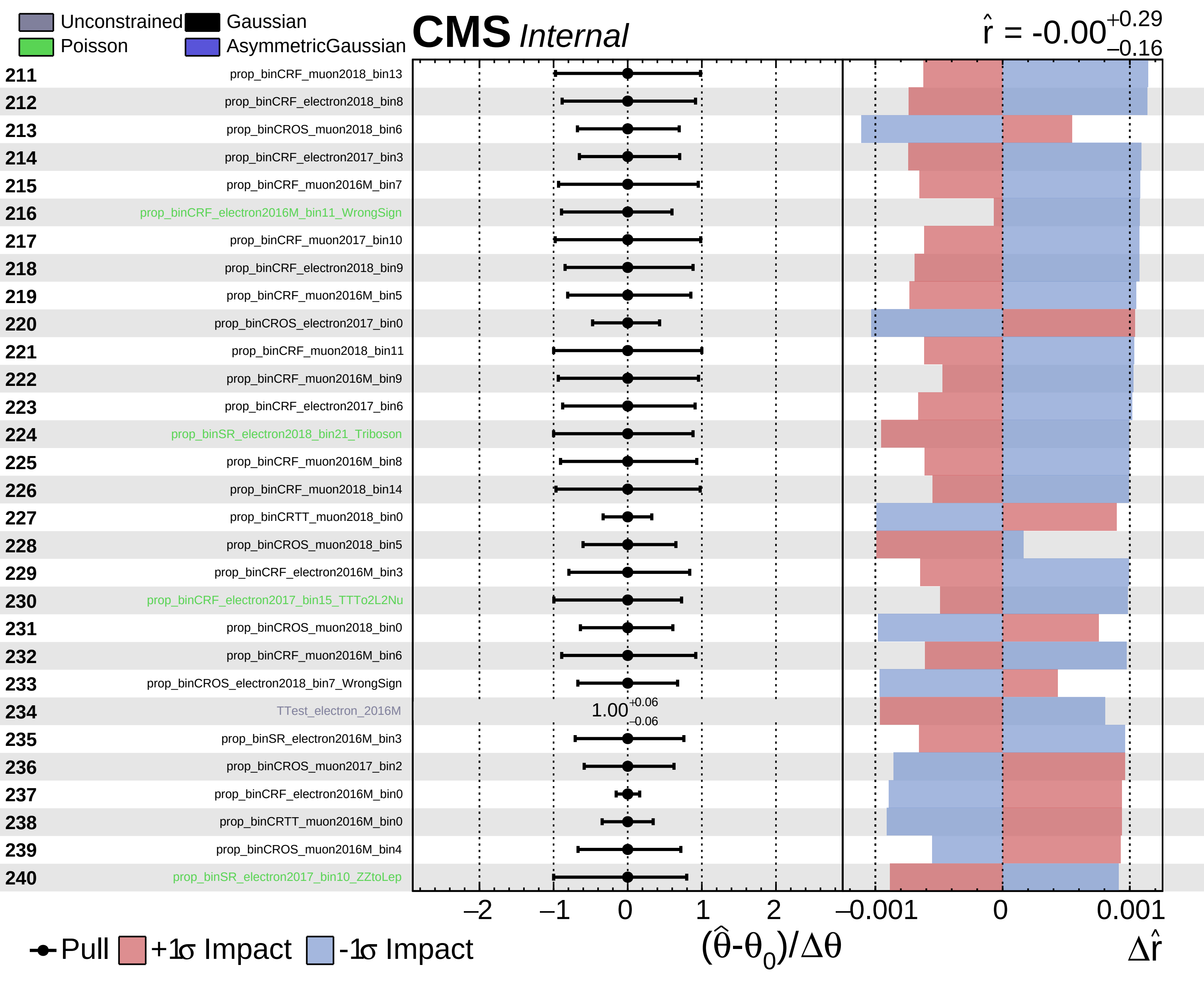
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.29}_{-0.16}$
121 prop_binCRF_muon2018_bin5
122 prop_binCROS_muon2018_bin4
123 prop_binSR_electron2016M_bin11
124 prop_binSR_electron2016M_bin17_TTTo2L2Nu
125 prop_binSR_muon2016M_bin2
126 prop_binCRF_muon2018_bin0
127 prop_binSR_electron2017_bin18
128 prop_binSR_electron2016M_bin16
129 prop_binSR_muon2017_bin21
130 prop_binSR_electron2017_bin9
131 prop_binSR_muon2017_bin17_ZZtoLep
132 prop_binSR_electron2017_bin5
133 prop_binSR_electron2017_bin8
134 pdf_total_Triboson
135 prop_binSR_electron2017_bin10_Triboson
136 prop_binSR_electron2017_bin6
137 prop_binSR_electron2017_bin4
138 FSR_VG
139 prop_binSR_electron2016M_bin18
140 prop_binSR_muon2016M_bin17_TTTo2L2Nu
141 prop_binSR_electron2016M_bin15
142 prop_binCRF_muon2018_bin4
143 prop_binSR_electron2017_bin3
144 prop_binSR_electron2016M_bin8
145 prop_binSR_muon2016M_bin1
146 prop_binCRF_electron2018_bin0
147 prop_binCRF_muon2018_bin8
148 prop_binSR_electron2016M_bin13
149 jes
150 prop_binCRF_muon2018_bin7
−2 −1 0 1 2
−0.002 0 0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



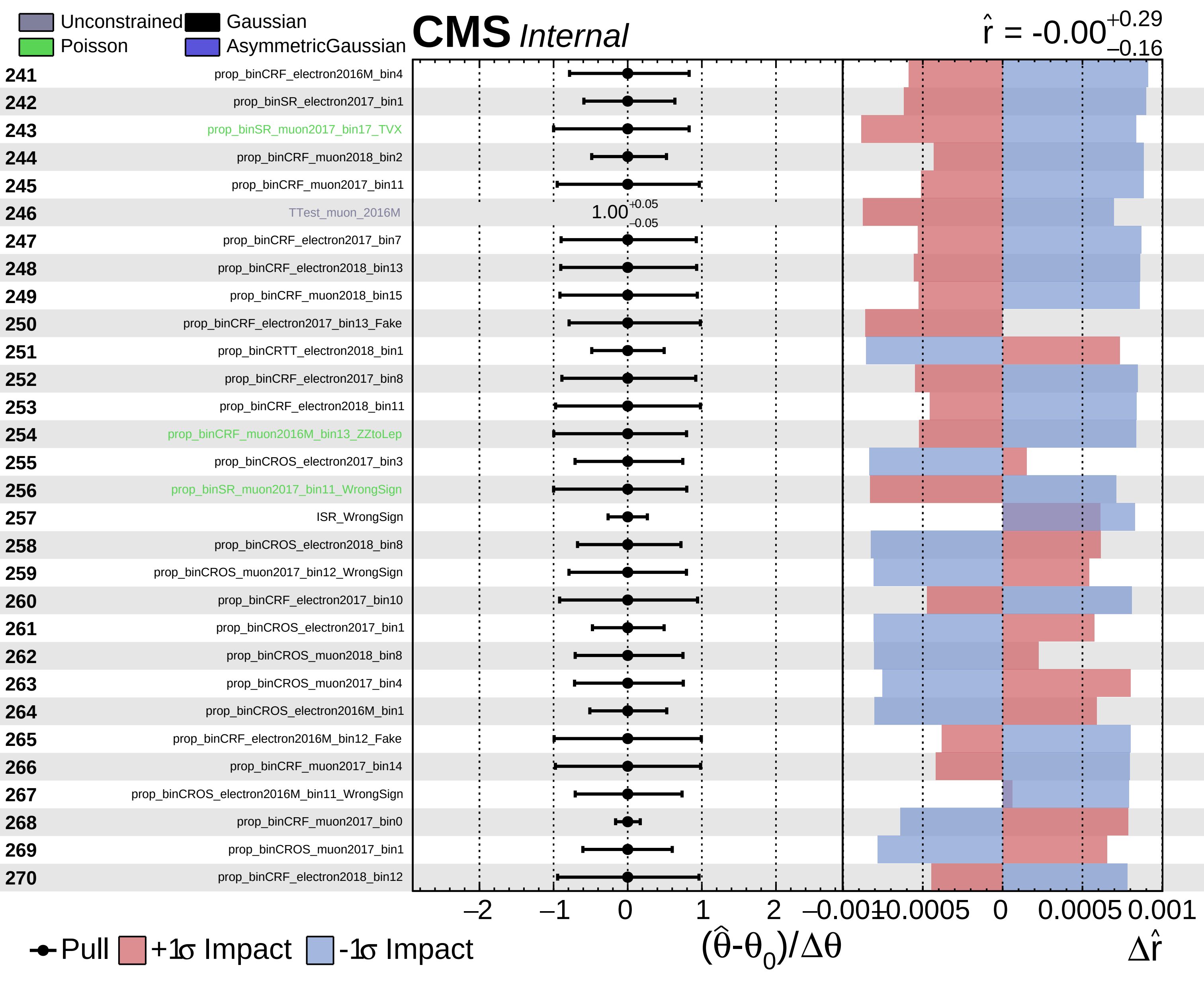
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.29}_{-0.16}$
151 prop_binSR_electron2016M_bin7
152 prop_binSR_electron2016M_bin19
153 prop_binCRF_muon2018_bin6
154 prop_binSR_muon2017_bin4
155 prop_binSR_electron2016M_bin14
156 prop_binSR_muon2016M_bin14_TTTo2L2Nu
157 prop_binSR_electron2016M_bin4
158 prop_binSR_electron2016M_bin10
159 pdf_total_TVX
160 prop_binCRF_muon2017_bin2
161 prop_binCROS_electron2018_bin1
162 prop_binSR_electron2016M_bin12
163 prop_binSR_electron2016M_bin9
164 prop_binSR_electron2016M_bin5
165 prop_binCRF_muon2017_bin5
166 prop_binCROS_muon2017_bin0
167 prop_binSR_muon2016M_bin17_ZZtoLep
168 prop_binSR_electron2016M_bin6
169 prop_binCRF_muon2018_bin9
170 prop_binCROS_electron2016M_bin0
171 prop_binCRF_muon2018_bin10
172 prop_binCRF_muon2017_bin4
173 prop_binCRF_electron2018_bin7
174 prop_binCRF_muon2018_bin12
175 QCDScale_VG
176 prop_binCROS_muon2018_bin12
177 mistag
178 prop_binCROS_muon2016M_bin1
179 prop_binCRF_muon2017_bin3
180 prop_binSR_electron2017_bin20_ZZtoLep
−2 −1 0 1 2
−0.002 −0.001 0 0.001 0.002
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

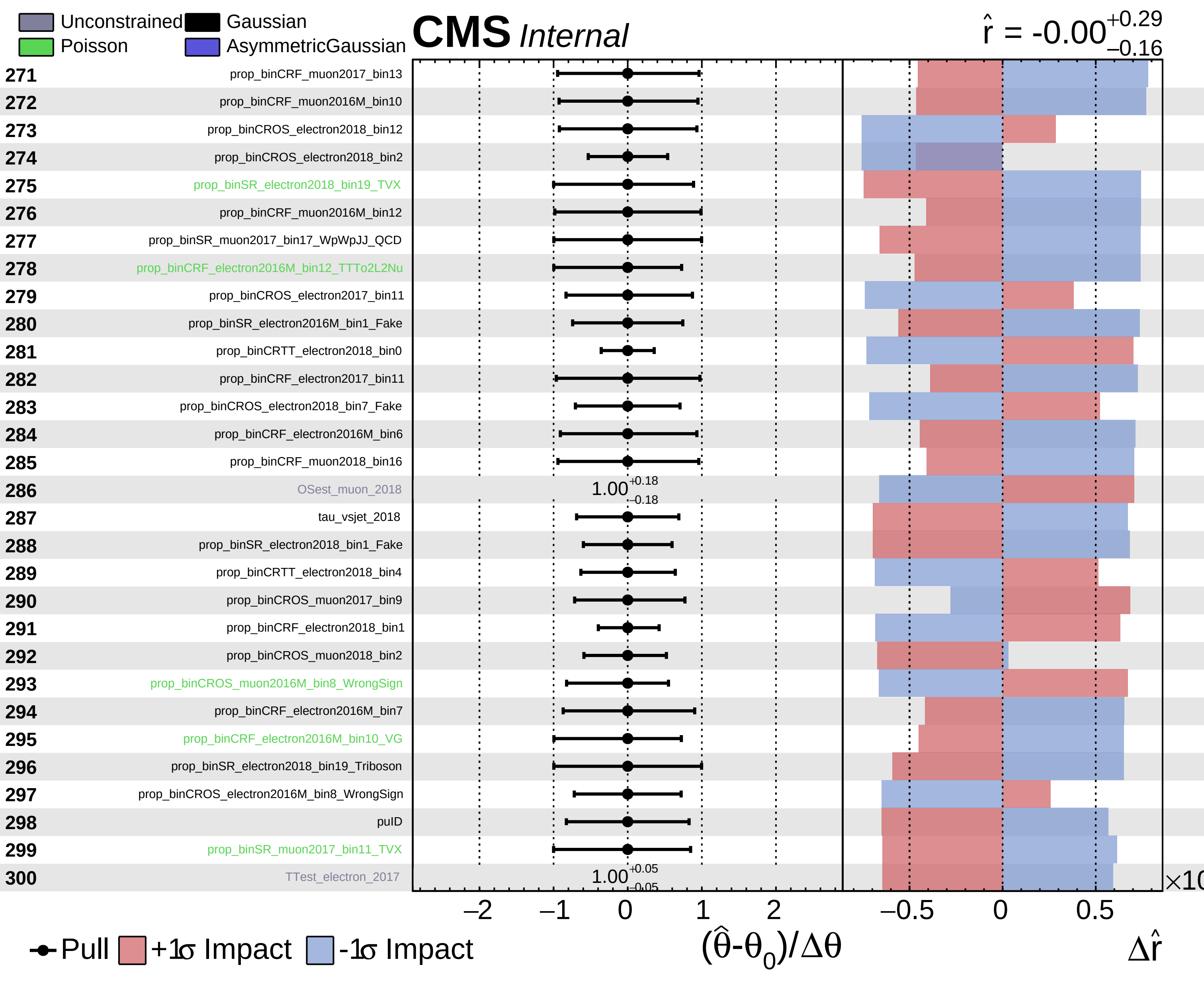


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.29}_{-0.16}$
181 prop_binSR_muon2016M_bin20_ZZtoLep
182 ISR_VG
183 prop_binCRF_electron2017_bin5
184 prop_binCROS_muon2018_bin1
185 prop_binCRF_electron2018_bin6
186 pu
187 prop_binCRF_muon2016M_bin3
188 prop_binSR_electron2018_bin2
189 prop_binCRF_electron2017_bin0
190 prop_binCRF_muon2017_bin15_Fake
191 prop_binCROS_electron2018_bin3
192 prop_binCRF_electron2018_bin10
193 prop_binSR_muon2017_bin11_ZZtoLep
194 prop_binCRF_electron2018_bin5
195 prop_binCROS_muon2018_bin3
196 tau_vsjet_2017
197 tau_vsjet_2016M
198 prop_binCROS_electron2018_bin0
199 OSest_muon_2016M
$1.00^{+0.27}_{-0.27}$
200 prop_binCRF_muon2017_bin6
201 prop_binCRF_muon2017_bin7
202 prop_binCRF_muon2016M_bin4
203 prop_binCROS_muon2016M_bin2
204 prop_binCRF_muon2017_bin9
205 prop_binCROS_electron2017_bin4
206 prop_binCRF_muon2017_bin8
207 prop_binCRF_electron2017_bin2
208 prop_binSR_electron2017_bin2
209 QCDScale_ZZtoLep
210 prop_binCRTT_electron2017_bin0
−2
−1
0
1
2
−0.001
0
0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

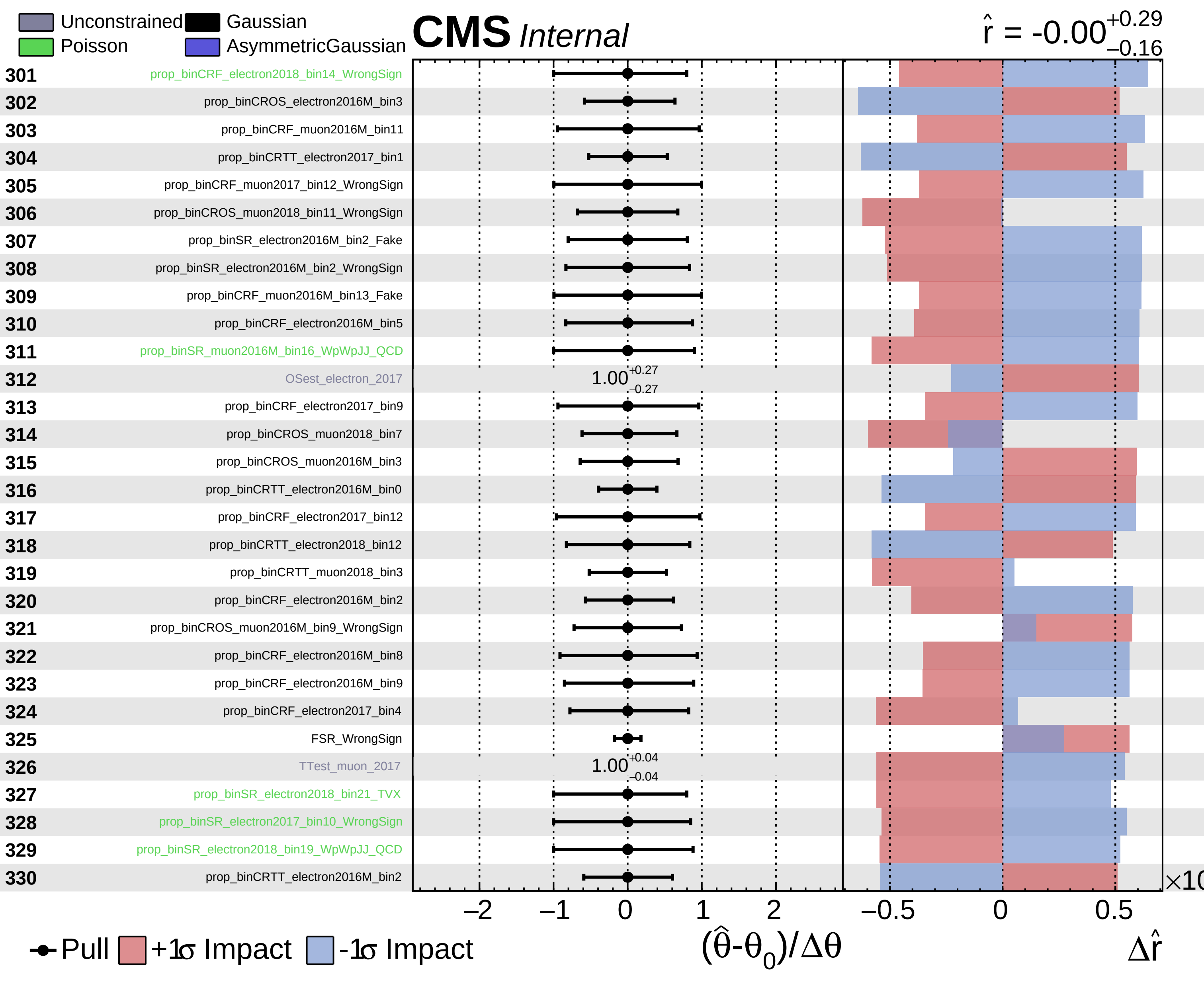


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.29}_{-0.16}$
211 prop_binCRF_muon2018_bin13
212 prop_binCRF_electron2018_bin8
213 prop_binCROS_muon2018_bin6
214 prop_binCRF_electron2017_bin3
215 prop_binCRF_muon2016M_bin7
216 prop_binCRF_electron2016M_bin11_WrongSign
217 prop_binCRF_muon2017_bin10
218 prop_binCRF_electron2018_bin9
219 prop_binCRF_muon2016M_bin5
220 prop_binCROS_electron2017_bin0
221 prop_binCRF_muon2018_bin11
222 prop_binCRF_muon2016M_bin9
223 prop_binCRF_electron2017_bin6
224 prop_binSR_electron2018_bin21_Triboson
225 prop_binCRF_muon2016M_bin8
226 prop_binCRF_muon2018_bin14
227 prop_binCRTT_muon2018_bin0
228 prop_binCROS_muon2018_bin5
229 prop_binCRF_electron2016M_bin3
230 prop_binCRF_electron2017_bin15_TTTo2L2Nu
231 prop_binCROS_muon2018_bin0
232 prop_binCRF_muon2016M_bin6
233 prop_binCROS_electron2018_bin7_WrongSign
234 TTest_electron_2016M
$1.00^{+0.06}_{-0.06}$
235 prop_binSR_electron2016M_bin3
236 prop_binCROS_muon2017_bin2
237 prop_binCRF_electron2016M_bin0
238 prop_binCRTT_muon2016M_bin0
239 prop_binCROS_muon2016M_bin4
240 prop_binSR_electron2017_bin10_ZZtoLep
−2 −1 0 1 2
−0.001 0 0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

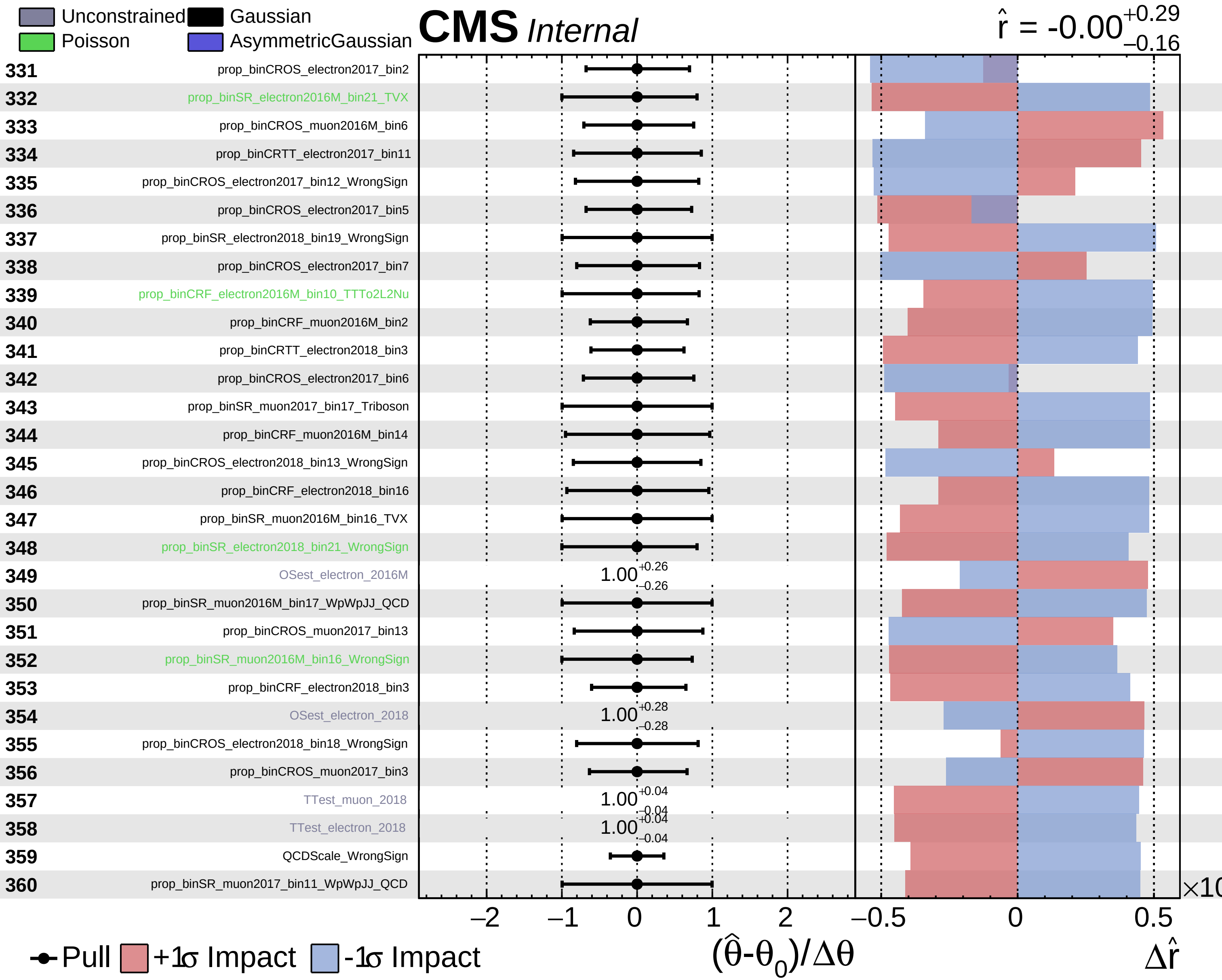

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.29}_{-0.16}$
241 prop_binCRF_electron2016M_bin4
242 prop_binSR_electron2017_bin1
243 prop_binSR_muon2017_bin17_TVX
244 prop_binCRF_muon2018_bin2
245 prop_binCRF_muon2017_bin11
246 TTest_muon_2016M
$1.00^{+0.05}_{-0.05}$
247 prop_binCRF_electron2017_bin7
248 prop_binCRF_electron2018_bin13
249 prop_binCRF_muon2018_bin15
250 prop_binCRF_electron2017_bin13_Fake
251 prop_binCRTT_electron2018_bin1
252 prop_binCRF_electron2017_bin8
253 prop_binCRF_electron2018_bin11
254 prop_binCRF_muon2016M_bin13_ZZtoLep
255 prop_binCROS_electron2017_bin3
256 prop_binSR_muon2017_bin11_WrongSign
257 ISR_WrongSign
258 prop_binCROS_electron2018_bin8
259 prop_binCROS_muon2017_bin12_WrongSign
260 prop_binCRF_electron2017_bin10
261 prop_binCROS_electron2017_bin1
262 prop_binCROS_muon2018_bin8
263 prop_binCROS_muon2017_bin4
264 prop_binCROS_electron2016M_bin1
265 prop_binCRF_electron2016M_bin12_Fake
266 prop_binCRF_muon2017_bin14
267 prop_binCROS_electron2016M_bin11_WrongSign
268 prop_binCRF_muon2017_bin0
269 prop_binCROS_muon2017_bin1
270 prop_binCRF_electron2018_bin12
−2 −1 0 1 2
−0.001 −0.0005 0 0.0005 0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

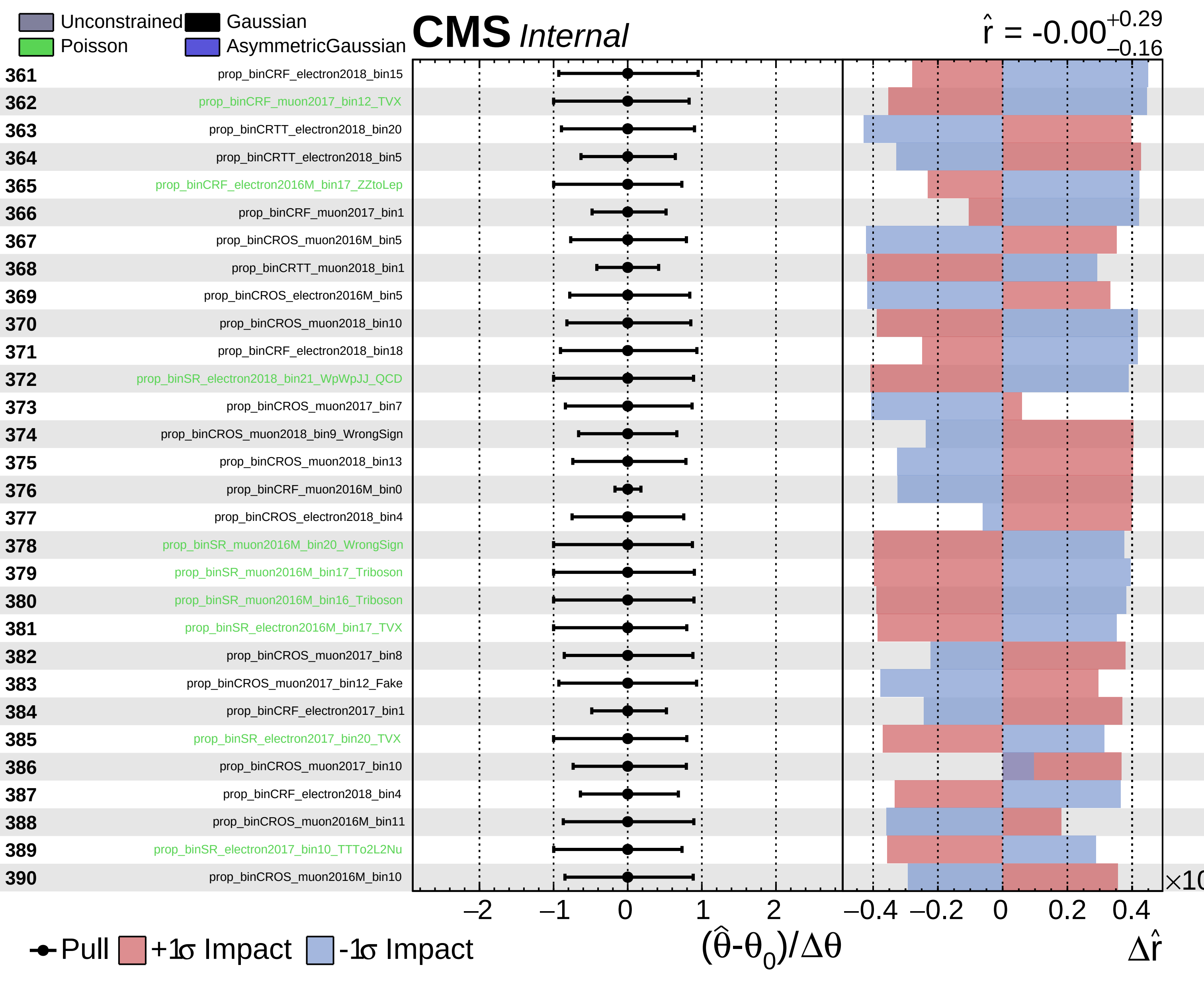


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.29}_{-0.16}$
271 prop_binCRF_muon2017_bin13
272 prop_binCRF_muon2016M_bin10
273 prop_binCROS_electron2018_bin12
274 prop_binCROS_electron2018_bin2
275 prop_binSR_electron2018_bin19_TVX
276 prop_binCRF_muon2016M_bin12
277 prop_binSR_muon2017_bin17_WpWpJJ_QCD
278 prop_binCRF_electron2016M_bin12_TTTo2L2Nu
279 prop_binCROS_electron2017_bin11
280 prop_binSR_electron2016M_bin1_Fake
281 prop_binCRTT_electron2018_bin0
282 prop_binCRF_electron2017_bin11
283 prop_binCROS_electron2018_bin7_Fake
284 prop_binCRF_electron2016M_bin6
285 prop_binCRF_muon2018_bin16
286 OSest_muon_2018
$1.00^{+0.18}_{-0.18}$
287 tau_vsjet_2018
288 prop_binSR_electron2018_bin1_Fake
289 prop_binCRTT_electron2018_bin4
290 prop_binCROS_muon2017_bin9
291 prop_binCRF_electron2018_bin1
292 prop_binCROS_muon2018_bin2
293 prop_binCROS_muon2016M_bin8_WrongSign
294 prop_binCRF_electron2016M_bin7
295 prop_binCRF_electron2016M_bin10_VG
296 prop_binSR_electron2018_bin19_Triboson
297 prop_binCROS_electron2016M_bin8_WrongSign
298 puID
299 prop_binSR_muon2017_bin11_TVX
300 TTest_electron_2017
$1.00^{+0.05}_{-0.05}$
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

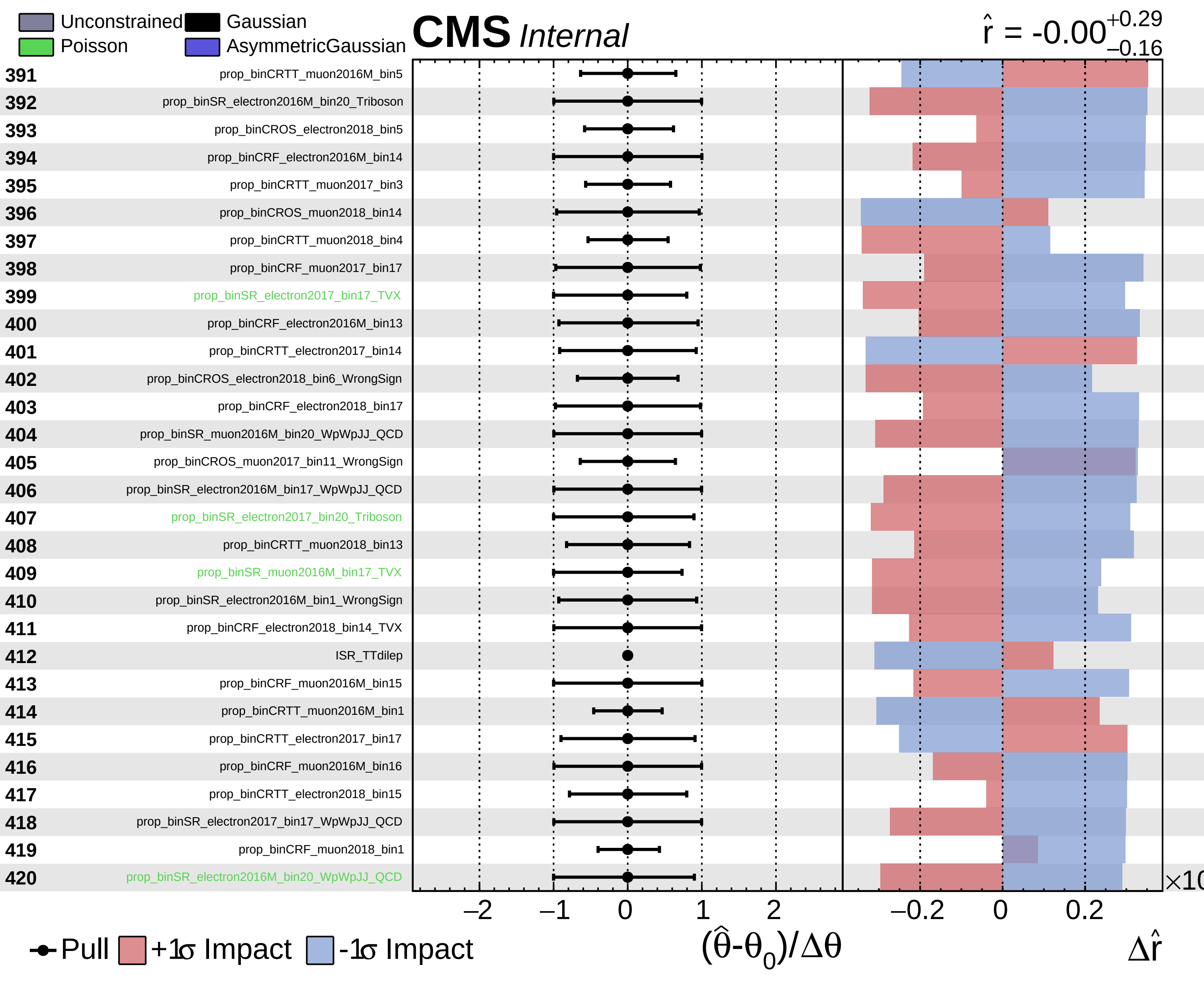


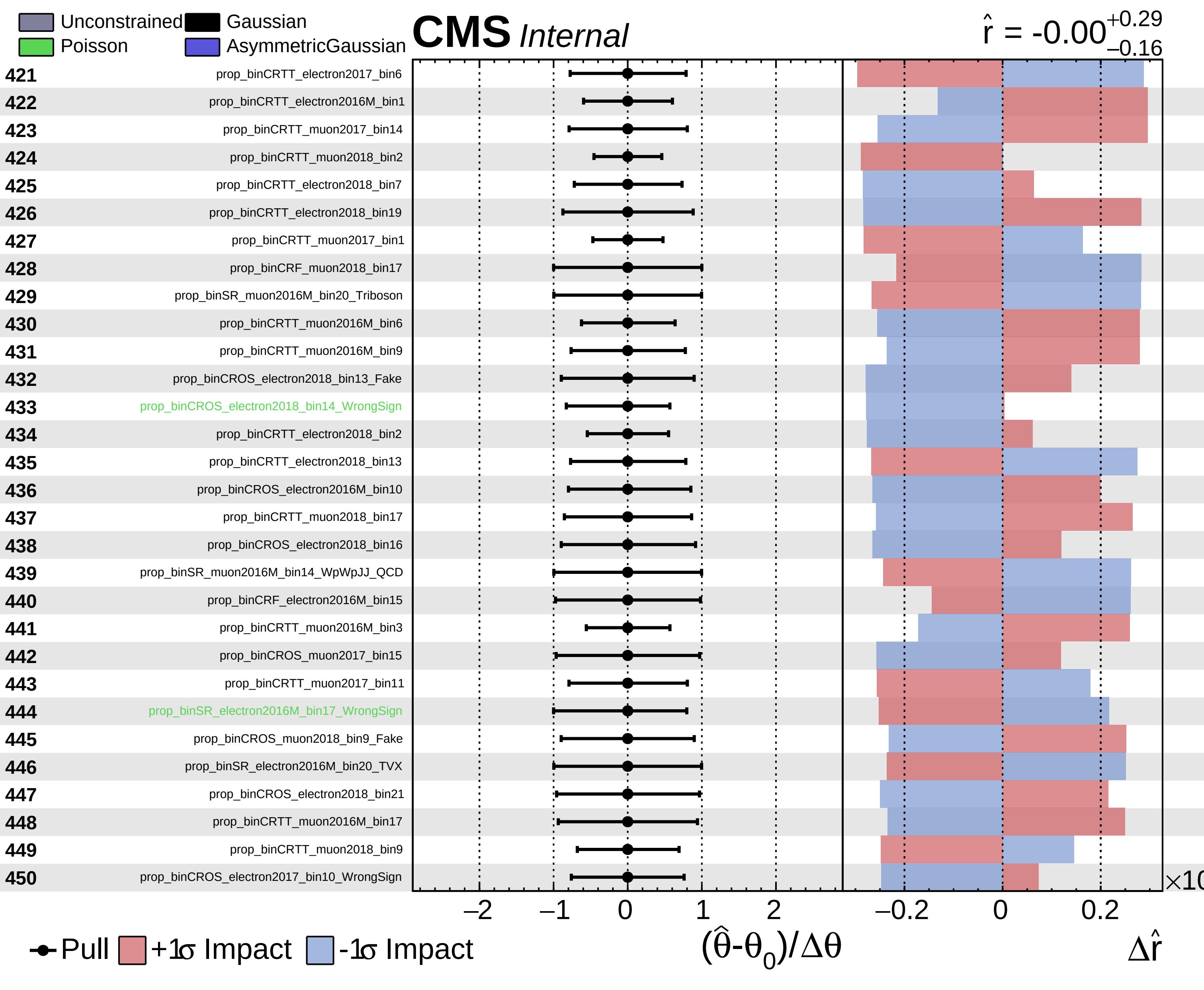
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.29}_{-0.16}$
301 prop_binCRF_electron2018_bin14_WrongSign
302 prop_binCROS_electron2016M_bin3
303 prop_binCRF_muon2016M_bin11
304 prop_binCRTT_electron2017_bin1
305 prop_binCRF_muon2017_bin12_WrongSign
306 prop_binCROS_muon2018_bin11_WrongSign
307 prop_binSR_electron2016M_bin2_Fake
308 prop_binSR_electron2016M_bin2_WrongSign
309 prop_binCRF_muon2016M_bin13_Fake
310 prop_binCRF_electron2016M_bin5
311 prop_binSR_muon2016M_bin16_WpWpJJ_QCD
312 OSest_electron_2017
$1.00^{+0.27}_{-0.27}$
313 prop_binCRF_electron2017_bin9
314 prop_binCROS_muon2018_bin7
315 prop_binCROS_muon2016M_bin3
316 prop_binCRTT_electron2016M_bin0
317 prop_binCRF_electron2017_bin12
318 prop_binCRTT_electron2018_bin12
319 prop_binCRTT_muon2018_bin3
320 prop_binCRF_electron2016M_bin2
321 prop_binCROS_muon2016M_bin9_WrongSign
322 prop_binCRF_electron2016M_bin8
323 prop_binCRF_electron2016M_bin9
324 prop_binCRF_electron2017_bin4
325 FSR_WrongSign
326 TTest_muon_2017
$1.00^{+0.04}_{-0.04}$
327 prop_binSR_electron2018_bin21_TVX
328 prop_binSR_electron2017_bin10_WrongSign
329 prop_binSR_electron2018_bin19_WpWpJJ_QCD
330 prop_binCRTT_electron2016M_bin2
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.29}_{-0.16}$
331 prop_binCROS_electron2017_bin2
332 prop_binSR_electron2016M_bin21_TVX
333 prop_binCROS_muon2016M_bin6
334 prop_binCRTT_electron2017_bin11
335 prop_binCROS_electron2017_bin12_WrongSign
336 prop_binCROS_electron2017_bin5
337 prop_binSR_electron2018_bin19_WrongSign
338 prop_binCROS_electron2017_bin7
339 prop_binCRF_electron2016M_bin10_TTTo2L2Nu
340 prop_binCRF_muon2016M_bin2
341 prop_binCRTT_electron2018_bin3
342 prop_binCROS_electron2017_bin6
343 prop_binSR_muon2017_bin17_Triboson
344 prop_binCRF_muon2016M_bin14
345 prop_binCROS_electron2018_bin13_WrongSign
346 prop_binCRF_electron2018_bin16
347 prop_binSR_muon2016M_bin16_TVX
348 prop_binSR_electron2018_bin21_WrongSign
349 OSest_electron_2016M $1.00^{+0.26}_{-0.26}$
350 prop_binSR_muon2016M_bin17_WpWpJJ_QCD
351 prop_binCROS_muon2017_bin13
352 prop_binSR_muon2016M_bin16_WrongSign
353 prop_binCRF_electron2018_bin3
354 OSest_electron_2018 $1.00^{+0.28}_{-0.28}$
355 prop_binCROS_electron2018_bin18_WrongSign
356 prop_binCROS_muon2017_bin3
357 TTest_muon_2018 $1.00^{+0.04}_{-0.04}$
358 TTest_electron_2018 $1.00^{+0.04}_{-0.04}$
359 QCDScale_WrongSign
360 prop_binSR_muon2017_bin11_WpWpJJ_QCD
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

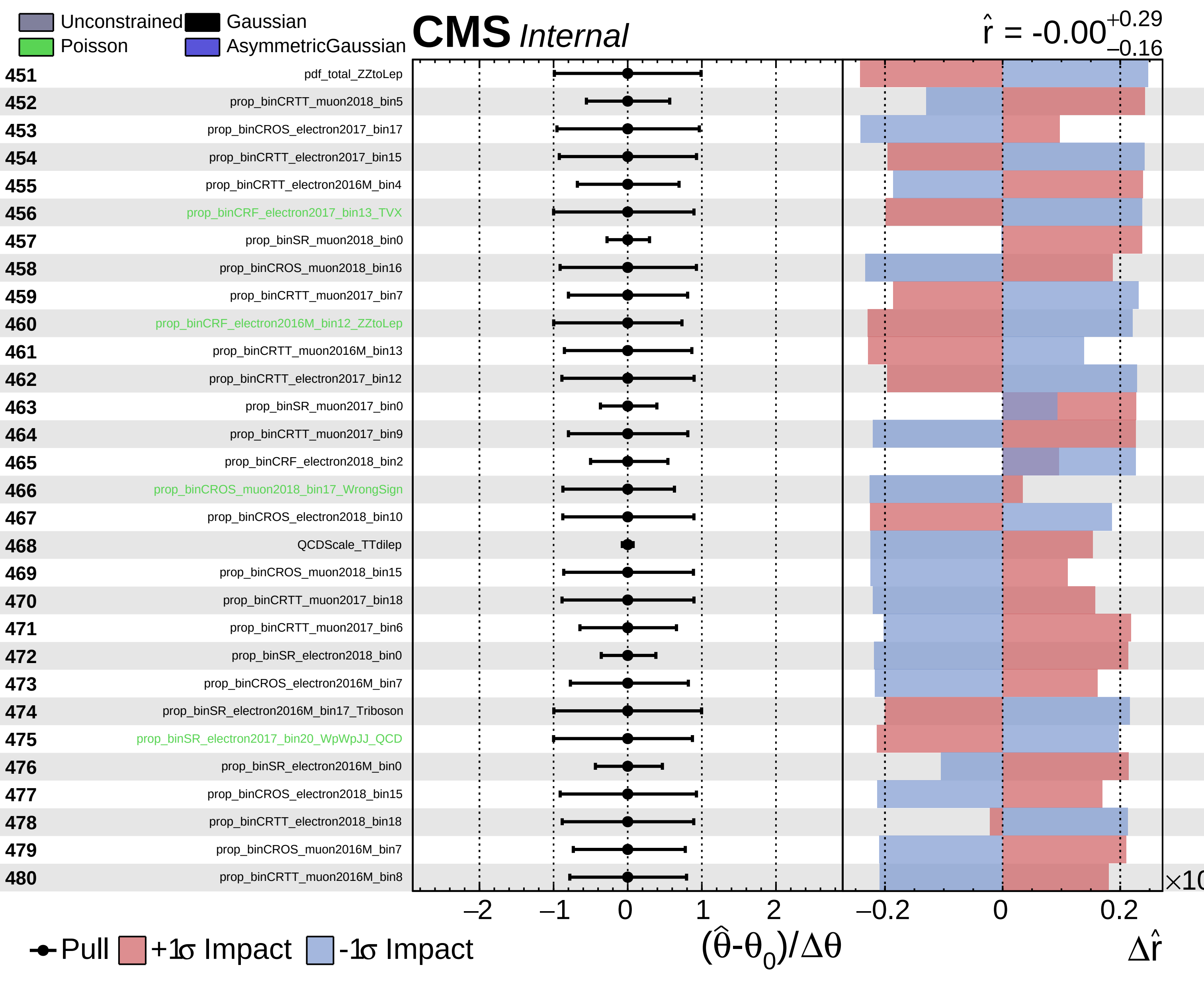


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = -0.00 +0.29 −0.16
361 prop_binCRF_electron2018_bin15
362 prop_binCRF_muon2017_bin12_TVX
363 prop_binCRTT_electron2018_bin20
364 prop_binCRTT_electron2018_bin5
365 prop_binCRF_electron2016M_bin17_ZZtoLep
366 prop_binCRF_muon2017_bin1
367 prop_binCROS_muon2016M_bin5
368 prop_binCRTT_muon2018_bin1
369 prop_binCROS_electron2016M_bin5
370 prop_binCROS_muon2018_bin10
371 prop_binCRF_electron2018_bin18
372 prop_binSR_electron2018_bin21_WpWpJJ_QCD
373 prop_binCROS_muon2017_bin7
374 prop_binCROS_muon2018_bin9_WrongSign
375 prop_binCROS_muon2018_bin13
376 prop_binCRF_muon2016M_bin0
377 prop_binCROS_electron2018_bin4
378 prop_binSR_muon2016M_bin20_WrongSign
379 prop_binSR_muon2016M_bin17_Triboson
380 prop_binSR_muon2016M_bin16_Triboson
381 prop_binSR_electron2016M_bin17_TVX
382 prop_binCROS_muon2017_bin8
383 prop_binCROS_muon2017_bin12_Fake
384 prop_binCRF_electron2017_bin1
385 prop_binSR_electron2017_bin20_TVX
386 prop_binCROS_muon2017_bin10
387 prop_binCRF_electron2018_bin4
388 prop_binCROS_muon2016M_bin11
389 prop_binSR_electron2017_bin10_TTTo2L2Nu
390 prop_binCROS_muon2016M_bin10
−2 −1 0 1 2
−0.4 −0.2 0 0.2 0.4
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

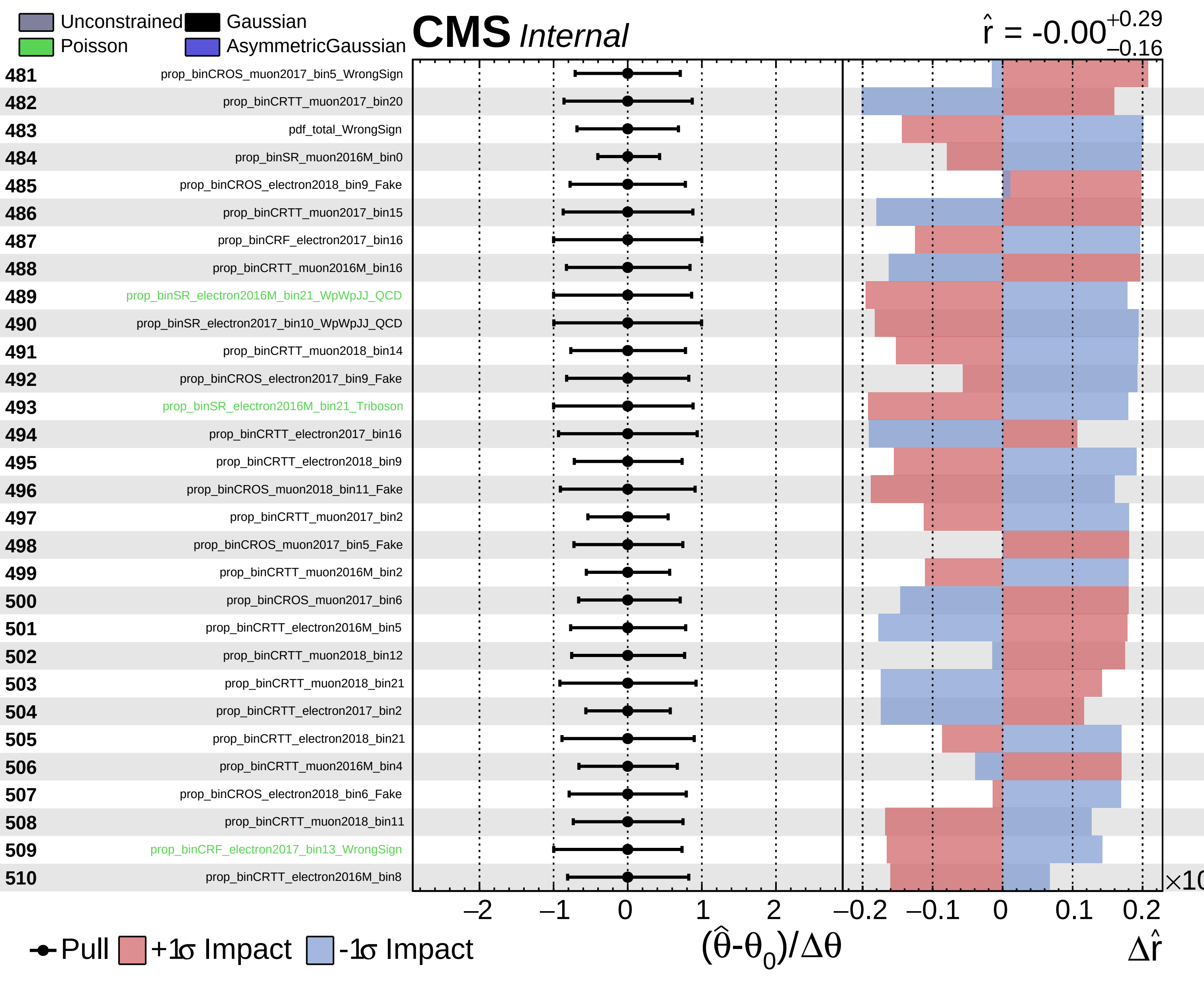

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.29}_{-0.16}$
391 prop_binCRTT_muon2016M_bin5
392 prop_binSR_electron2016M_bin20_Triboson
393 prop_binCROS_electron2018_bin5
394 prop_binCRF_electron2016M_bin14
395 prop_binCRTT_muon2017_bin3
396 prop_binCROS_muon2018_bin14
397 prop_binCRTT_muon2018_bin4
398 prop_binCRF_muon2017_bin17
399 prop_binSR_electron2017_bin17_TVX
400 prop_binCRF_electron2016M_bin13
401 prop_binCRTT_electron2017_bin14
402 prop_binCROS_electron2018_bin6_WrongSign
403 prop_binCRF_electron2018_bin17
404 prop_binSR_muon2016M_bin20_WpWpJJ_QCD
405 prop_binCROS_muon2017_bin11_WrongSign
406 prop_binSR_electron2016M_bin17_WpWpJJ_QCD
407 prop_binSR_electron2017_bin20_Triboson
408 prop_binCRTT_muon2018_bin13
409 prop_binSR_muon2016M_bin17_TVX
410 prop_binSR_electron2016M_bin1_WrongSign
411 prop_binCRF_electron2018_bin14_TVX
412 ISR_TTdilep
413 prop_binCRF_muon2016M_bin15
414 prop_binCRTT_muon2016M_bin1
415 prop_binCRTT_electron2017_bin17
416 prop_binCRF_muon2016M_bin16
417 prop_binCRTT_electron2018_bin15
418 prop_binSR_electron2017_bin17_WpWpJJ_QCD
419 prop_binCRF_muon2018_bin1
420 prop_binSR_electron2016M_bin20_WpWpJJ_QCD
−2
−1
0
1
2
−0.2
0
0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



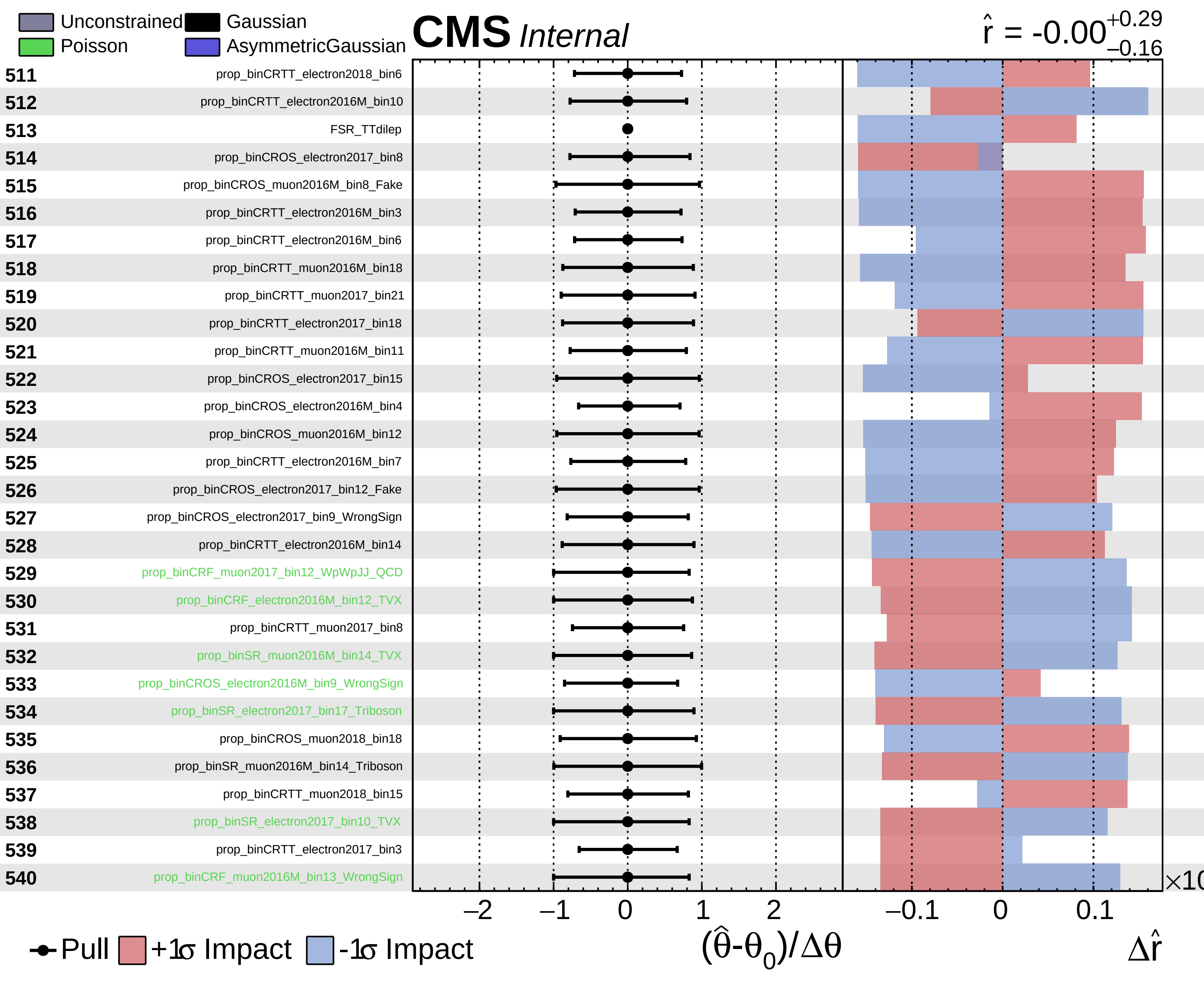
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = -0.00 +0.29 −0.16
421 prop_binCRTT_electron2017_bin6
422 prop_binCRTT_electron2016M_bin1
423 prop_binCRTT_muon2017_bin14
424 prop_binCRTT_muon2018_bin2
425 prop_binCRTT_electron2018_bin7
426 prop_binCRTT_electron2018_bin19
427 prop_binCRTT_muon2017_bin1
428 prop_binCRF_muon2018_bin17
429 prop_binSR_muon2016M_bin20_Triboson
430 prop_binCRTT_muon2016M_bin6
431 prop_binCRTT_muon2016M_bin9
432 prop_binCROS_electron2018_bin13_Fake
433 prop_binCROS_electron2018_bin14_WrongSign
434 prop_binCRTT_electron2018_bin2
435 prop_binCRTT_electron2018_bin13
436 prop_binCROS_electron2016M_bin10
437 prop_binCRTT_muon2018_bin17
438 prop_binCROS_electron2018_bin16
439 prop_binSR_muon2016M_bin14_WpWpJJ_QCD
440 prop_binCRF_electron2016M_bin15
441 prop_binCRTT_muon2016M_bin3
442 prop_binCROS_muon2017_bin15
443 prop_binCRTT_muon2017_bin11
444 prop_binSR_electron2016M_bin17_WrongSign
445 prop_binCROS_muon2018_bin9_Fake
446 prop_binSR_electron2016M_bin20_TVX
447 prop_binCROS_electron2018_bin21
448 prop_binCRTT_muon2016M_bin17
449 prop_binCRTT_muon2018_bin9
450 prop_binCROS_electron2017_bin10_WrongSign
×10
−2 −1 0 1 2
−0.2 0 0.2
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δr̂

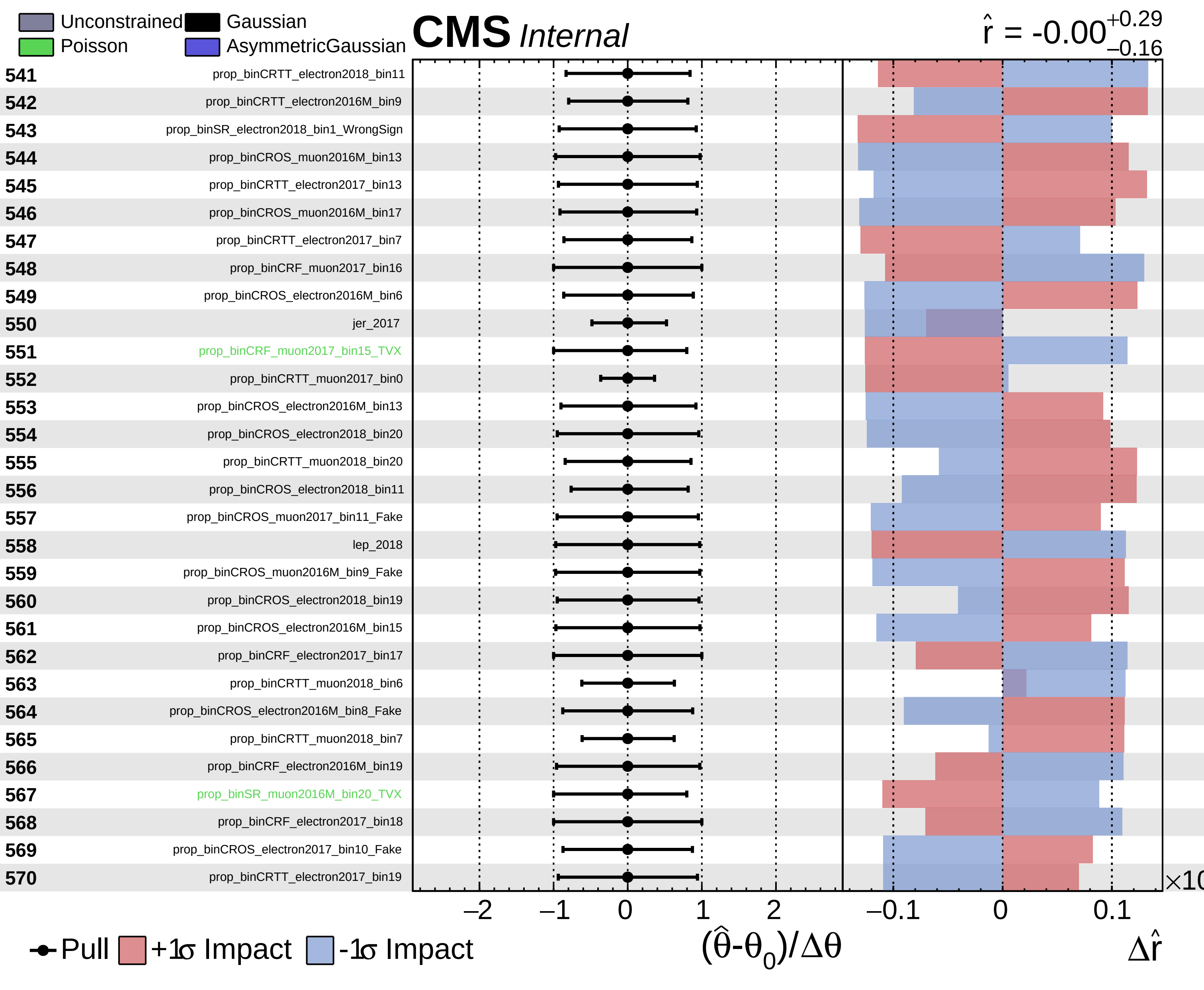


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.29}_{-0.16}$
451 pdf_total_ZZtoLep
452 prop_binCRTT_muon2018_bin5
453 prop_binCROS_electron2017_bin17
454 prop_binCRTT_electron2017_bin15
455 prop_binCRTT_electron2016M_bin4
456 prop_binCRF_electron2017_bin13_TVX
457 prop_binSR_muon2018_bin0
458 prop_binCROS_muon2018_bin16
459 prop_binCRTT_muon2017_bin7
460 prop_binCRF_electron2016M_bin12_ZZtoLep
461 prop_binCRTT_muon2016M_bin13
462 prop_binCRTT_electron2017_bin12
463 prop_binSR_muon2017_bin0
464 prop_binCRTT_muon2017_bin9
465 prop_binCRF_electron2018_bin2
466 prop_binCROS_muon2018_bin17_WrongSign
467 prop_binCROS_electron2018_bin10
468 QCDScale_TTdilep
469 prop_binCROS_muon2018_bin15
470 prop_binCRTT_muon2017_bin18
471 prop_binCRTT_muon2017_bin6
472 prop_binSR_electron2018_bin0
473 prop_binCROS_electron2016M_bin7
474 prop_binSR_electron2016M_bin17_Triboson
475 prop_binSR_electron2017_bin20_WpWpJJ_QCD
476 prop_binSR_electron2016M_bin0
477 prop_binCROS_electron2018_bin15
478 prop_binCRTT_electron2018_bin18
479 prop_binCROS_muon2016M_bin7
480 prop_binCRTT_muon2016M_bin8
−2 −1 0 1 2
−0.2 0 0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

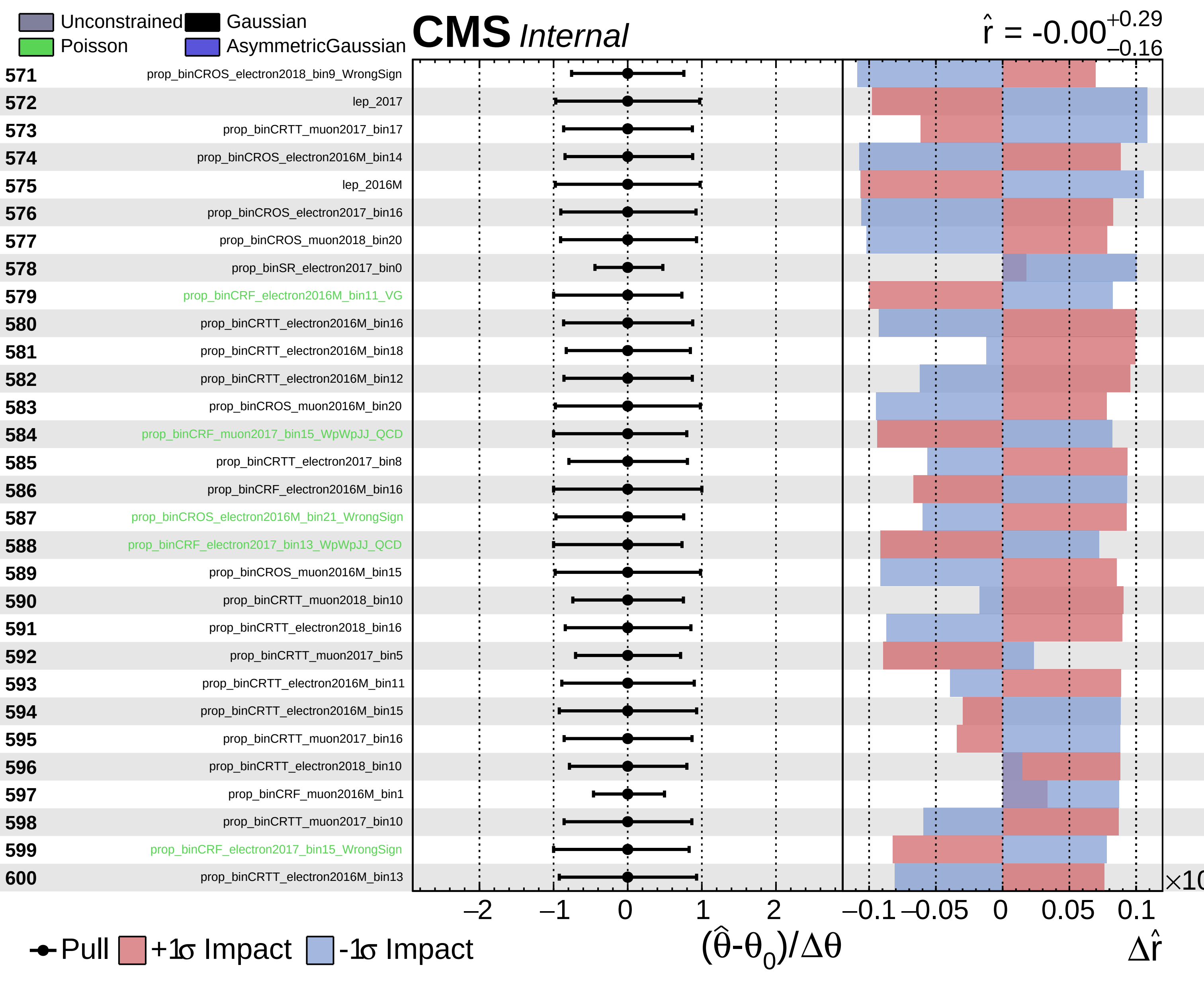


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = -0.00 +0.29 −0.16
481 prop_binCROS_muon2017_bin5_WrongSign
482 prop_binCRTT_muon2017_bin20
483 pdf_total_WrongSign
484 prop_binSR_muon2016M_bin0
485 prop_binCROS_electron2018_bin9_Fake
486 prop_binCRTT_muon2017_bin15
487 prop_binCRF_electron2017_bin16
488 prop_binCRTT_muon2016M_bin16
489 prop_binSR_electron2016M_bin21_WpWpJJ_QCD
490 prop_binSR_electron2017_bin10_WpWpJJ_QCD
491 prop_binCRTT_muon2018_bin14
492 prop_binCROS_electron2017_bin9_Fake
493 prop_binSR_electron2016M_bin21_Triboson
494 prop_binCRTT_electron2017_bin16
495 prop_binCRTT_electron2018_bin9
496 prop_binCROS_muon2018_bin11_Fake
497 prop_binCRTT_muon2017_bin2
498 prop_binCROS_muon2017_bin5_Fake
499 prop_binCRTT_muon2016M_bin2
500 prop_binCROS_muon2017_bin6
501 prop_binCRTT_electron2016M_bin5
502 prop_binCRTT_muon2018_bin12
503 prop_binCRTT_muon2018_bin21
504 prop_binCRTT_electron2017_bin2
505 prop_binCRTT_electron2018_bin21
506 prop_binCRTT_muon2016M_bin4
507 prop_binCROS_electron2018_bin6_Fake
508 prop_binCRTT_muon2018_bin11
509 prop_binCRF_electron2017_bin13_WrongSign
510 prop_binCRTT_electron2016M_bin8
−2 −1 0 1 2
−0.2 −0.1 0 0.1 0.2
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂

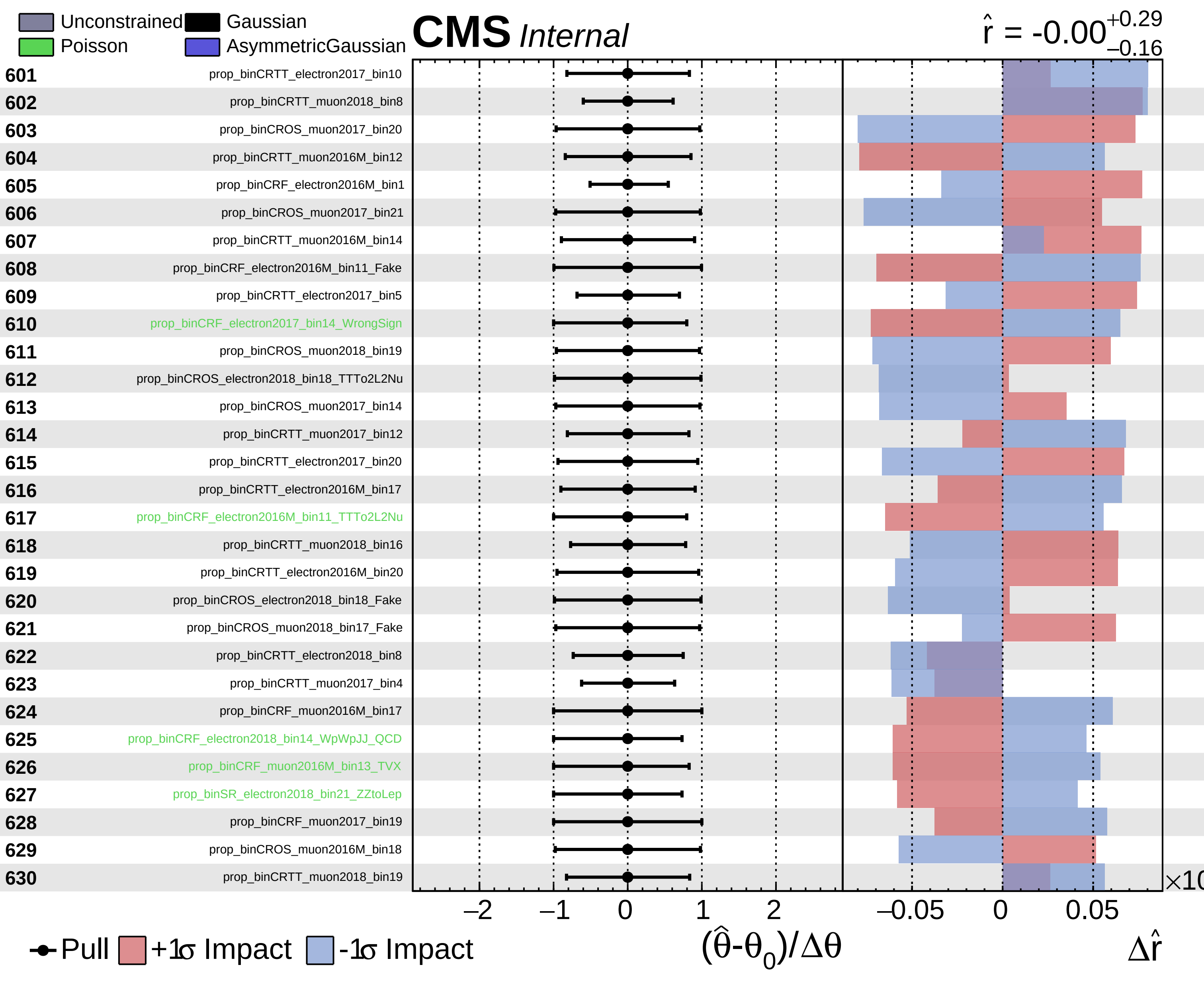

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.29}_{-0.16}$
511 prop_binCRTT_electron2018_bin6
512 prop_binCRTT_electron2016M_bin10
513 FSR_TTdilep
514 prop_binCROS_electron2017_bin8
515 prop_binCROS_muon2016M_bin8_Fake
516 prop_binCRTT_electron2016M_bin3
517 prop_binCRTT_electron2016M_bin6
518 prop_binCRTT_muon2016M_bin18
519 prop_binCRTT_muon2017_bin21
520 prop_binCRTT_electron2017_bin18
521 prop_binCRTT_muon2016M_bin11
522 prop_binCROS_electron2017_bin15
523 prop_binCROS_electron2016M_bin4
524 prop_binCROS_muon2016M_bin12
525 prop_binCRTT_electron2016M_bin7
526 prop_binCROS_electron2017_bin12_Fake
527 prop_binCROS_electron2017_bin9_WrongSign
528 prop_binCRTT_electron2016M_bin14
529 prop_binCRF_muon2017_bin12_WpWpJJ_QCD
530 prop_binCRF_electron2016M_bin12_TVX
531 prop_binCRTT_muon2017_bin8
532 prop_binSR_muon2016M_bin14_TVX
533 prop_binCROS_electron2016M_bin9_WrongSign
534 prop_binSR_electron2017_bin17_Triboson
535 prop_binCROS_muon2018_bin18
536 prop_binSR_muon2016M_bin14_Triboson
537 prop_binCRTT_muon2018_bin15
538 prop_binSR_electron2017_bin10_TVX
539 prop_binCRTT_electron2017_bin3
540 prop_binCRF_muon2016M_bin13_WrongSign
−2 −1 0 1 2
−0.1 0 0.1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



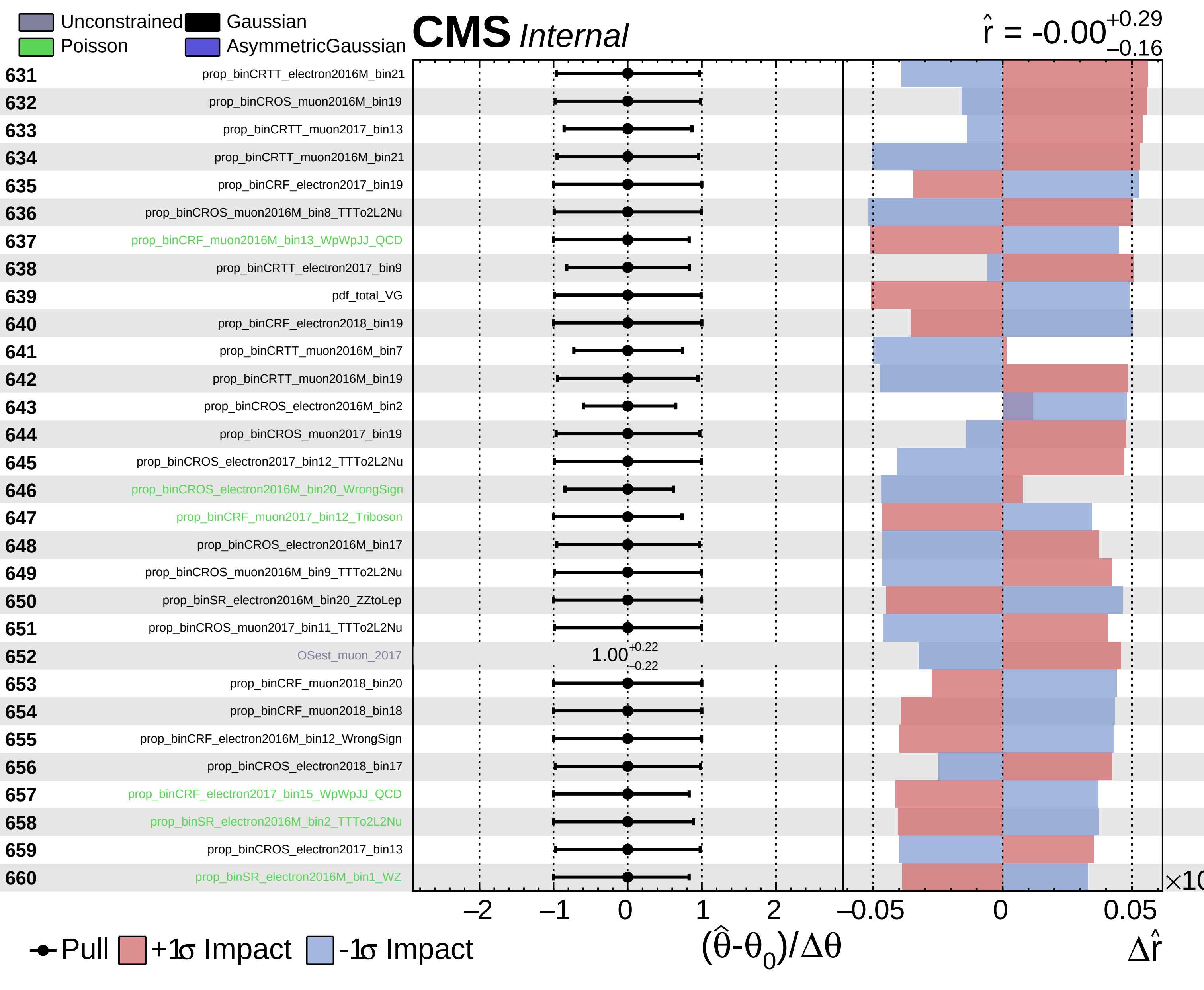
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.29}_{-0.16}$
541 prop_binCRTT_electron2018_bin11
542 prop_binCRTT_electron2016M_bin9
543 prop_binSR_electron2018_bin1_WrongSign
544 prop_binCROS_muon2016M_bin13
545 prop_binCRTT_electron2017_bin13
546 prop_binCROS_muon2016M_bin17
547 prop_binCRTT_electron2017_bin7
548 prop_binCRF_muon2017_bin16
549 prop_binCROS_electron2016M_bin6
550 jer_2017
551 prop_binCRF_muon2017_bin15_TVX
552 prop_binCRTT_muon2017_bin0
553 prop_binCROS_electron2016M_bin13
554 prop_binCROS_electron2018_bin20
555 prop_binCRTT_muon2018_bin20
556 prop_binCROS_electron2018_bin11
557 prop_binCROS_muon2017_bin11_Fake
558 lep_2018
559 prop_binCROS_muon2016M_bin9_Fake
560 prop_binCROS_electron2018_bin19
561 prop_binCROS_electron2016M_bin15
562 prop_binCRF_electron2017_bin17
563 prop_binCRTT_muon2018_bin6
564 prop_binCROS_electron2016M_bin8_Fake
565 prop_binCRTT_muon2018_bin7
566 prop_binCRF_electron2016M_bin19
567 prop_binSR_muon2016M_bin20_TVX
568 prop_binCRF_electron2017_bin18
569 prop_binCROS_electron2017_bin10_Fake
570 prop_binCRTT_electron2017_bin19
−2 −1 0 1 2
−0.1 0 0.1
×10
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



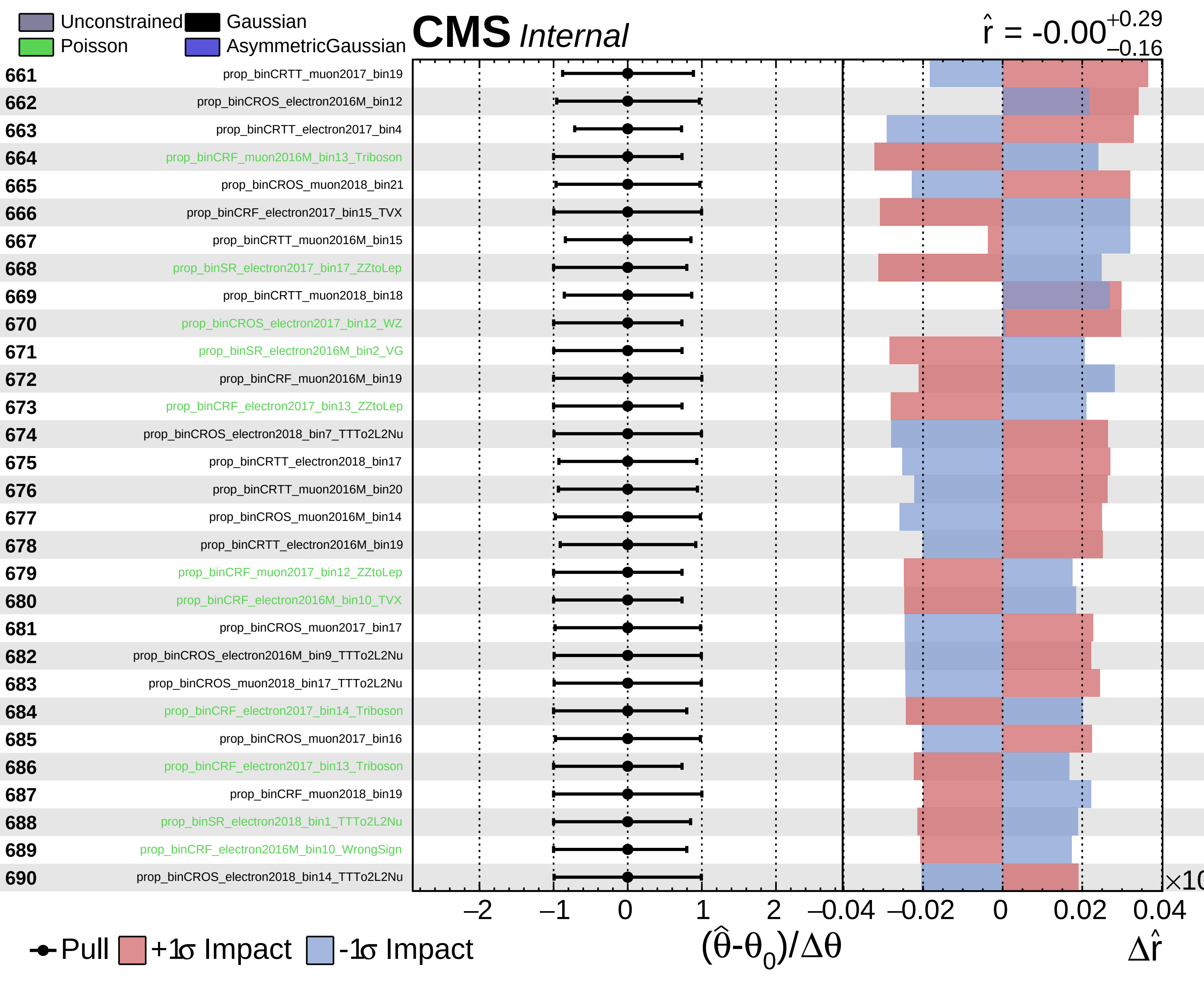
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.29}_{-0.16}$
571 prop_binCROS_electron2018_bin9_WrongSign
572 lep_2017
573 prop_binCRTT_muon2017_bin17
574 prop_binCROS_electron2016M_bin14
575 lep_2016M
576 prop_binCROS_electron2017_bin16
577 prop_binCROS_muon2018_bin20
578 prop_binSR_electron2017_bin0
579 prop_binCRF_electron2016M_bin11_VG
580 prop_binCRTT_electron2016M_bin16
581 prop_binCRTT_electron2016M_bin18
582 prop_binCRTT_electron2016M_bin12
583 prop_binCROS_muon2016M_bin20
584 prop_binCRF_muon2017_bin15_WpWpJJ_QCD
585 prop_binCRTT_electron2017_bin8
586 prop_binCRF_electron2016M_bin16
587 prop_binCROS_electron2016M_bin21_WrongSign
588 prop_binCRF_electron2017_bin13_WpWpJJ_QCD
589 prop_binCROS_muon2016M_bin15
590 prop_binCRTT_muon2018_bin10
591 prop_binCRTT_electron2018_bin16
592 prop_binCRTT_muon2017_bin5
593 prop_binCRTT_electron2016M_bin11
594 prop_binCRTT_electron2016M_bin15
595 prop_binCRTT_muon2017_bin16
596 prop_binCRTT_electron2018_bin10
597 prop_binCRF_muon2016M_bin1
598 prop_binCRTT_muon2017_bin10
599 prop_binCRF_electron2017_bin15_WrongSign
600 prop_binCRTT_electron2016M_bin13
−2 −1 0 1 2
−0.1 −0.05 0 0.05 0.1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



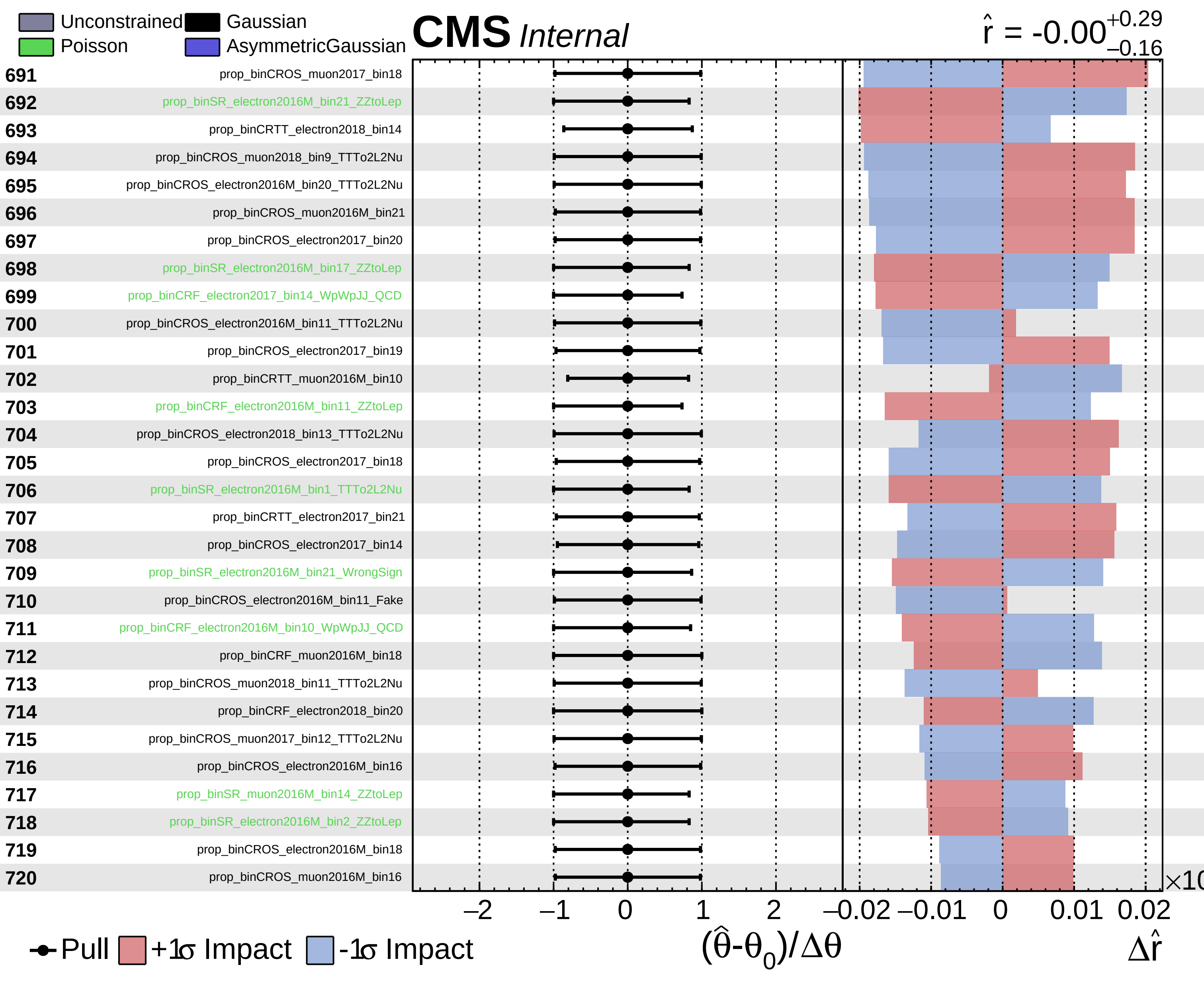
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = -0.00 +0.29 −0.16
601 prop_binCRTT_electron2017_bin10
602 prop_binCRTT_muon2018_bin8
603 prop_binCROS_muon2017_bin20
604 prop_binCRTT_muon2016M_bin12
605 prop_binCRF_electron2016M_bin1
606 prop_binCROS_muon2017_bin21
607 prop_binCRTT_muon2016M_bin14
608 prop_binCRF_electron2016M_bin11_Fake
609 prop_binCRTT_electron2017_bin5
610 prop_binCRF_electron2017_bin14_WrongSign
611 prop_binCROS_muon2018_bin19
612 prop_binCROS_electron2018_bin18_TTTo2L2Nu
613 prop_binCROS_muon2017_bin14
614 prop_binCRTT_muon2017_bin12
615 prop_binCRTT_electron2017_bin20
616 prop_binCRTT_electron2016M_bin17
617 prop_binCRF_electron2016M_bin11_TTTo2L2Nu
618 prop_binCRTT_muon2018_bin16
619 prop_binCRTT_electron2016M_bin20
620 prop_binCROS_electron2018_bin18_Fake
621 prop_binCROS_muon2018_bin17_Fake
622 prop_binCRTT_electron2018_bin8
623 prop_binCRTT_muon2017_bin4
624 prop_binCRF_muon2016M_bin17
625 prop_binCRF_electron2018_bin14_WpWpJJ_QCD
626 prop_binCRF_muon2016M_bin13_TVX
627 prop_binSR_electron2018_bin21_ZZtoLep
628 prop_binCRF_muon2017_bin19
629 prop_binCROS_muon2016M_bin18
630 prop_binCRTT_muon2018_bin19
−2 −1 0 1 2
−0.05 0 0.05
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂



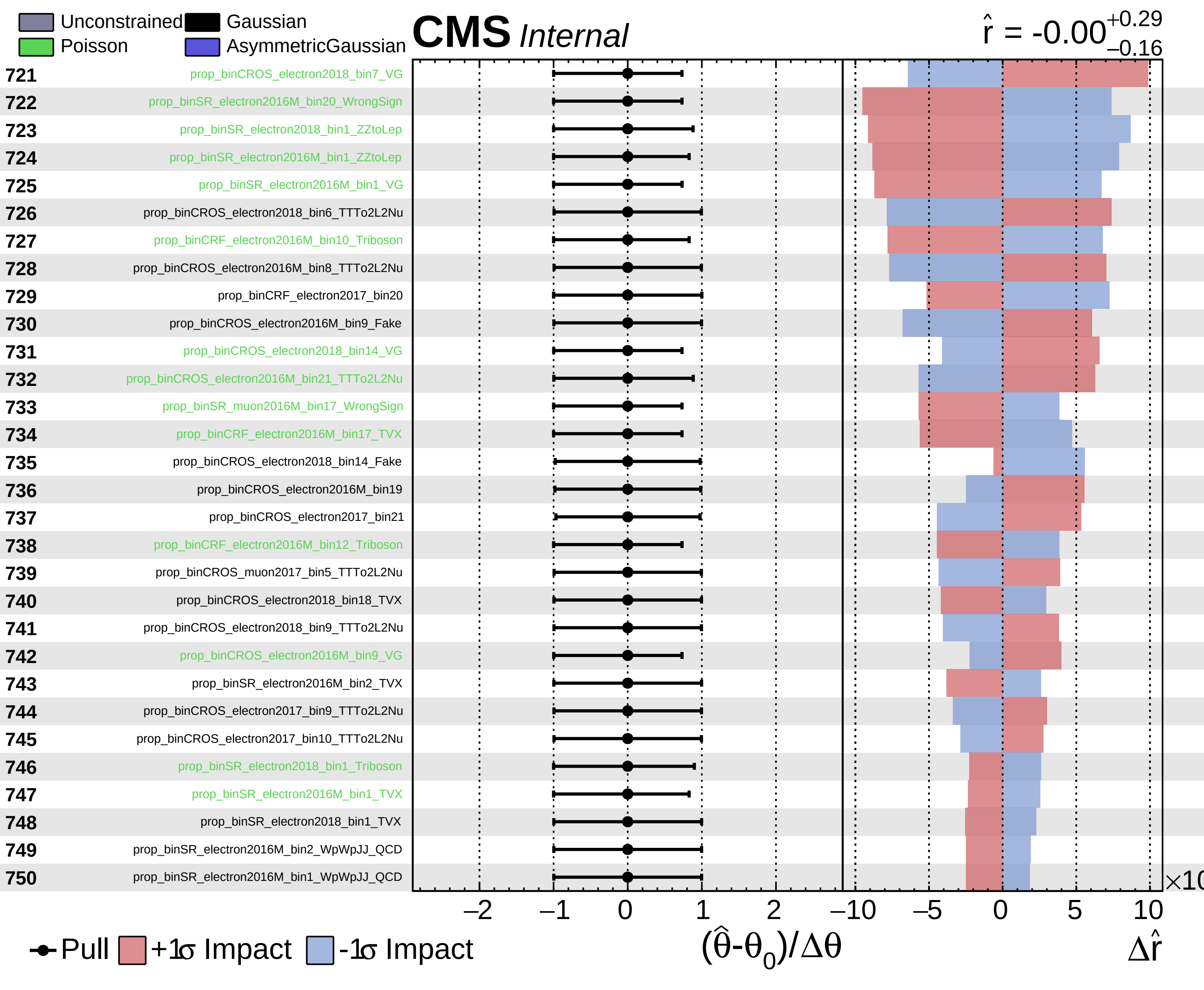
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.29}_{-0.16}$
631 prop_binCRTT_electron2016M_bin21
632 prop_binCROS_muon2016M_bin19
633 prop_binCRTT_muon2017_bin13
634 prop_binCRTT_muon2016M_bin21
635 prop_binCRF_electron2017_bin19
636 prop_binCROS_muon2016M_bin8_TTTo2L2Nu
637 prop_binCRF_muon2016M_bin13_WpWpJJ_QCD
638 prop_binCRTT_electron2017_bin9
639 pdf_total_VG
640 prop_binCRF_electron2018_bin19
641 prop_binCRTT_muon2016M_bin7
642 prop_binCRTT_muon2016M_bin19
643 prop_binCROS_electron2016M_bin2
644 prop_binCROS_muon2017_bin19
645 prop_binCROS_electron2017_bin12_TTTo2L2Nu
646 prop_binCROS_electron2016M_bin20_WrongSign
647 prop_binCRF_muon2017_bin12_Triboson
648 prop_binCROS_electron2016M_bin17
649 prop_binCROS_muon2016M_bin9_TTTo2L2Nu
650 prop_binSR_electron2016M_bin20_ZZtoLep
651 prop_binCROS_muon2017_bin11_TTTo2L2Nu
652 OSest_muon_2017 $1.00^{+0.22}_{-0.22}$
653 prop_binCRF_muon2018_bin20
654 prop_binCRF_muon2018_bin18
655 prop_binCRF_electron2016M_bin12_WrongSign
656 prop_binCROS_electron2018_bin17
657 prop_binCRF_electron2017_bin15_WpWpJJ_QCD
658 prop_binSR_electron2016M_bin2_TTTo2L2Nu
659 prop_binCROS_electron2017_bin13
660 prop_binSR_electron2016M_bin1_WZ
−2 −1 0 1 2
−0.05 0 0.05
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

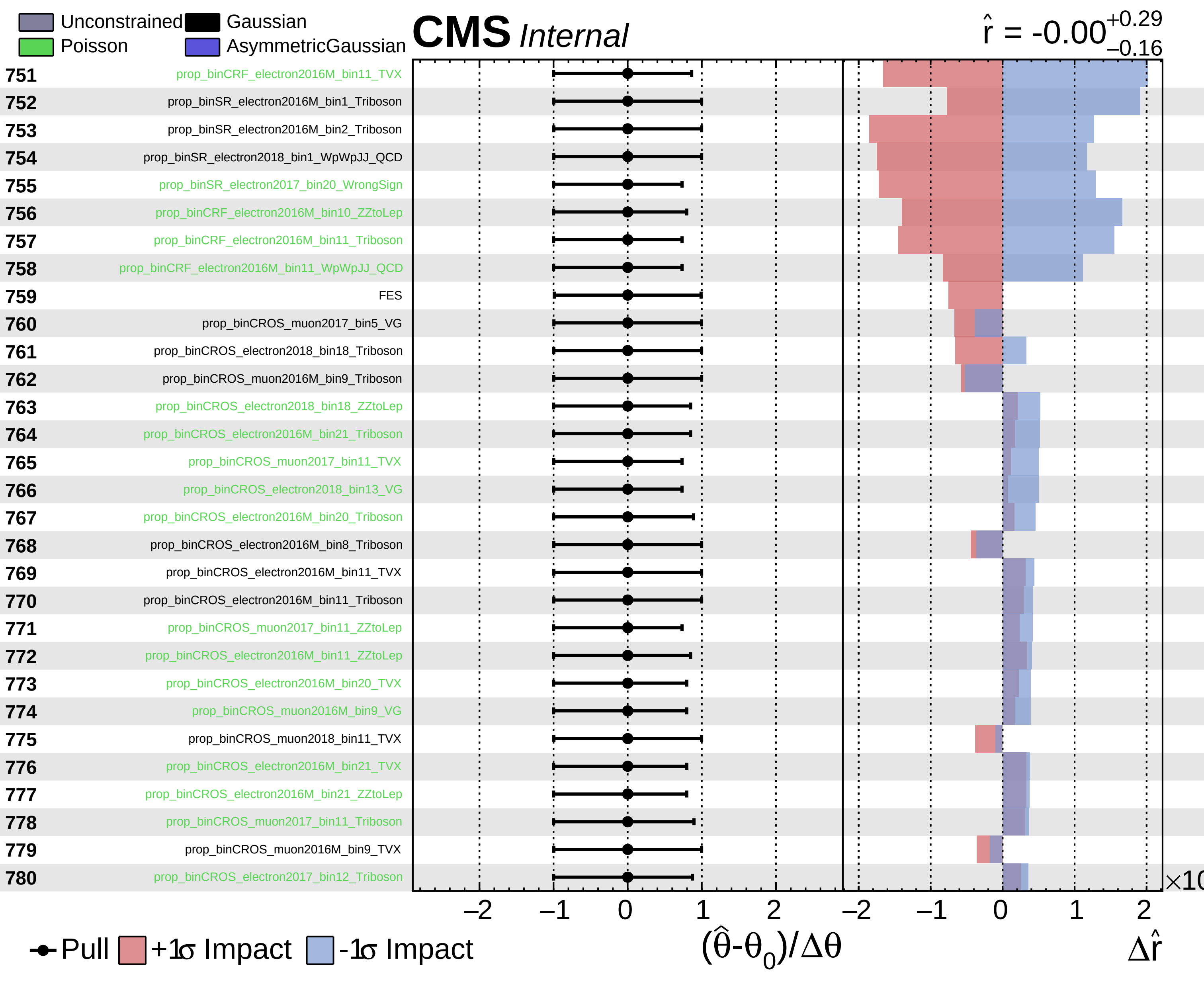


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.29}_{-0.16}$
661 prop_binCRTT_muon2017_bin19
662 prop_binCROS_electron2016M_bin12
663 prop_binCRTT_electron2017_bin4
664 prop_binCRF_muon2016M_bin13_Triboson
665 prop_binCROS_muon2018_bin21
666 prop_binCRF_electron2017_bin15_TVX
667 prop_binCRTT_muon2016M_bin15
668 prop_binSR_electron2017_bin17_ZZtoLep
669 prop_binCRTT_muon2018_bin18
670 prop_binCROS_electron2017_bin12_WZ
671 prop_binSR_electron2016M_bin2_VG
672 prop_binCRF_muon2016M_bin19
673 prop_binCRF_electron2017_bin13_ZZtoLep
674 prop_binCROS_electron2018_bin7_TTTo2L2Nu
675 prop_binCRTT_electron2018_bin17
676 prop_binCRTT_muon2016M_bin20
677 prop_binCROS_muon2016M_bin14
678 prop_binCRTT_electron2016M_bin19
679 prop_binCRF_muon2017_bin12_ZZtoLep
680 prop_binCRF_electron2016M_bin10_TVX
681 prop_binCROS_muon2017_bin17
682 prop_binCROS_electron2016M_bin9_TTTo2L2Nu
683 prop_binCROS_muon2018_bin17_TTTo2L2Nu
684 prop_binCRF_electron2017_bin14_Triboson
685 prop_binCROS_muon2017_bin16
686 prop_binCRF_electron2017_bin13_Triboson
687 prop_binCRF_muon2018_bin19
688 prop_binSR_electron2018_bin1_TTTo2L2Nu
689 prop_binCRF_electron2016M_bin10_WrongSign
690 prop_binCROS_electron2018_bin14_TTTo2L2Nu
−2 −1 0 1 2
−0.04 −0.02 0 0.02 0.04
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

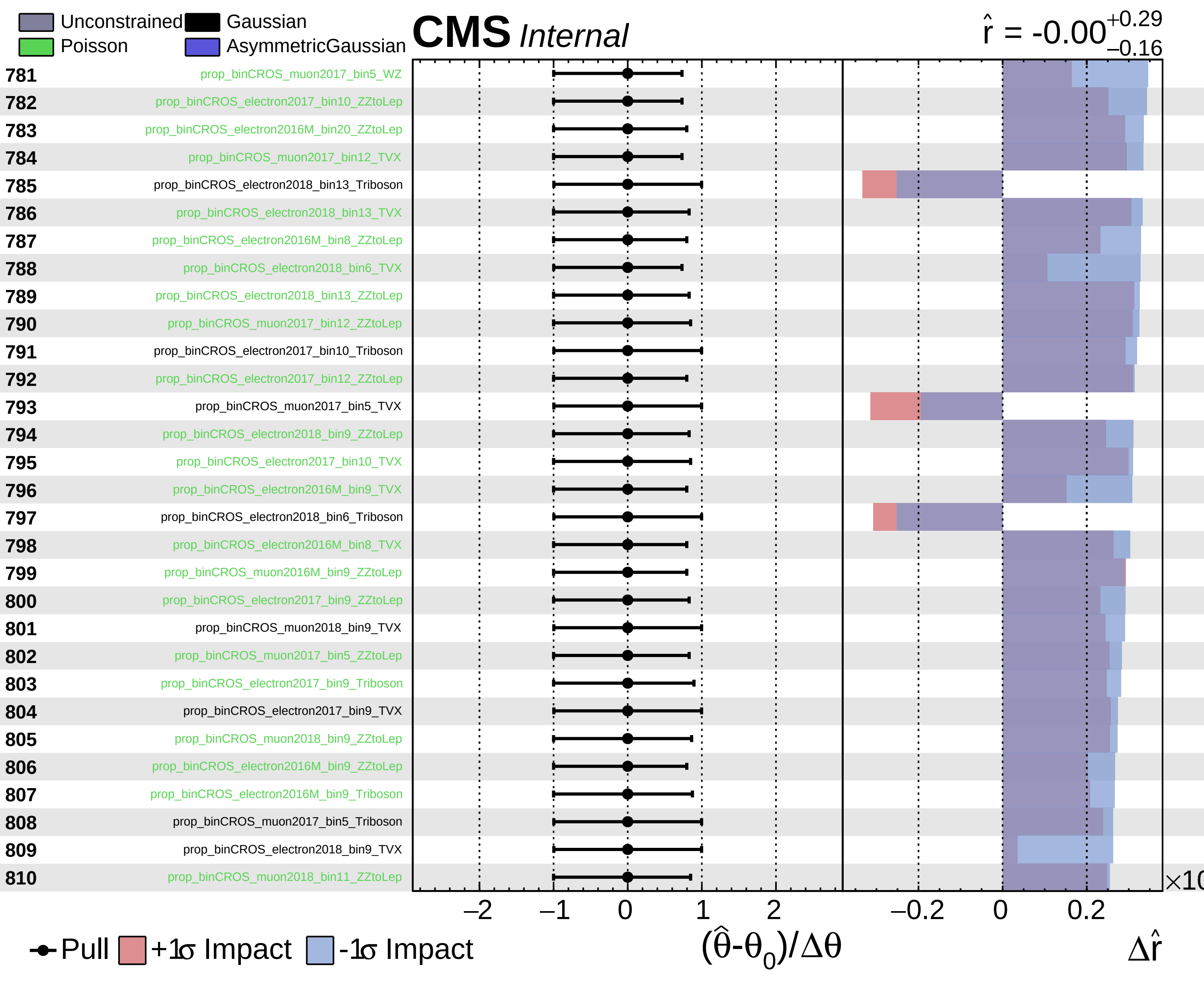


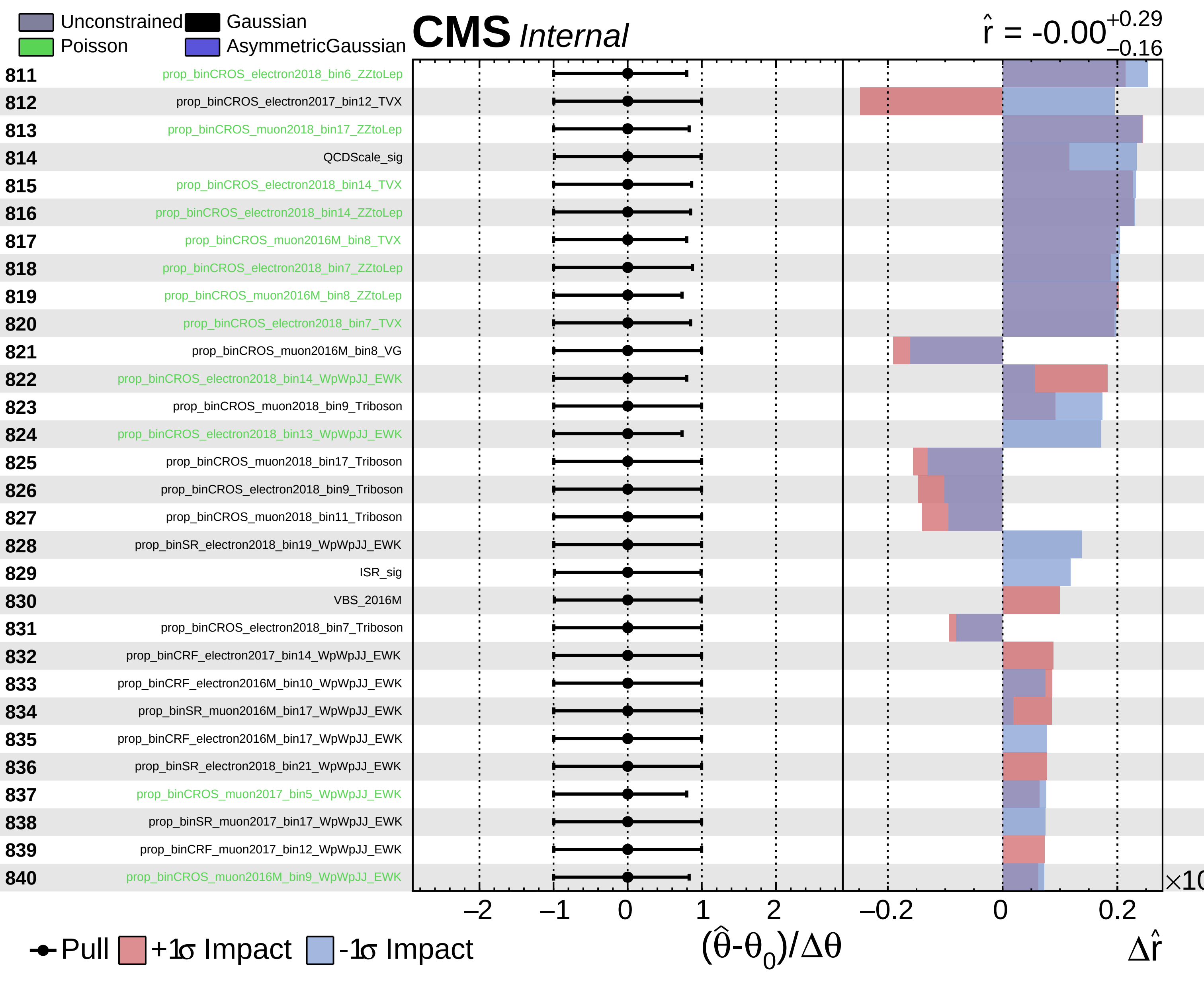
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.29}_{-0.16}$
691 prop_binCROS_muon2017_bin18
692 prop_binSR_electron2016M_bin21_ZZtoLep
693 prop_binCRTT_electron2018_bin14
694 prop_binCROS_muon2018_bin9_TTTo2L2Nu
695 prop_binCROS_electron2016M_bin20_TTTo2L2Nu
696 prop_binCROS_muon2016M_bin21
697 prop_binCROS_electron2017_bin20
698 prop_binSR_electron2016M_bin17_ZZtoLep
699 prop_binCRF_electron2017_bin14_WpWpJJ_QCD
700 prop_binCROS_electron2016M_bin11_TTTo2L2Nu
701 prop_binCROS_electron2017_bin19
702 prop_binCRTT_muon2016M_bin10
703 prop_binCRF_electron2016M_bin11_ZZtoLep
704 prop_binCROS_electron2018_bin13_TTTo2L2Nu
705 prop_binCROS_electron2017_bin18
706 prop_binSR_electron2016M_bin1_TTTo2L2Nu
707 prop_binCRTT_electron2017_bin21
708 prop_binCROS_electron2017_bin14
709 prop_binSR_electron2016M_bin21_WrongSign
710 prop_binCROS_electron2016M_bin11_Fake
711 prop_binCRF_electron2016M_bin10_WpWpJJ_QCD
712 prop_binCRF_muon2016M_bin18
713 prop_binCROS_muon2018_bin11_TTTo2L2Nu
714 prop_binCRF_electron2018_bin20
715 prop_binCROS_muon2017_bin12_TTTo2L2Nu
716 prop_binCROS_electron2016M_bin16
717 prop_binSR_muon2016M_bin14_ZZtoLep
718 prop_binSR_electron2016M_bin2_ZZtoLep
719 prop_binCROS_electron2016M_bin18
720 prop_binCROS_muon2016M_bin16
−2 −1 0 1 2
−0.02 −0.01 0 0.01 0.02
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.29}_{-0.16}$
721 prop_binCROS_electron2018_bin7_VG
722 prop_binSR_electron2016M_bin20_WrongSign
723 prop_binSR_electron2018_bin1_ZZtoLep
724 prop_binSR_electron2016M_bin1_ZZtoLep
725 prop_binSR_electron2016M_bin1_VG
726 prop_binCROS_electron2018_bin6_TTTo2L2Nu
727 prop_binCRF_electron2016M_bin10_Triboson
728 prop_binCROS_electron2016M_bin8_TTTo2L2Nu
729 prop_binCRF_electron2017_bin20
730 prop_binCROS_electron2016M_bin9_Fake
731 prop_binCROS_electron2018_bin14_VG
732 prop_binCROS_electron2016M_bin21_TTTo2L2Nu
733 prop_binSR_muon2016M_bin17_WrongSign
734 prop_binCRF_electron2016M_bin17_TVX
735 prop_binCROS_electron2018_bin14_Fake
736 prop_binCROS_electron2016M_bin19
737 prop_binCROS_electron2017_bin21
738 prop_binCRF_electron2016M_bin12_Triboson
739 prop_binCROS_muon2017_bin5_TTTo2L2Nu
740 prop_binCROS_electron2018_bin18_TVX
741 prop_binCROS_electron2018_bin9_TTTo2L2Nu
742 prop_binCROS_electron2016M_bin9_VG
743 prop_binSR_electron2016M_bin2_TVX
744 prop_binCROS_electron2017_bin9_TTTo2L2Nu
745 prop_binCROS_electron2017_bin10_TTTo2L2Nu
746 prop_binSR_electron2018_bin1_Triboson
747 prop_binSR_electron2016M_bin1_TVX
748 prop_binSR_electron2018_bin1_TVX
749 prop_binSR_electron2016M_bin2_WpWpJJ_QCD
750 prop_binSR_electron2016M_bin1_WpWpJJ_QCD
−2
−1
0
1
2
−10
−5
0
5
10
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

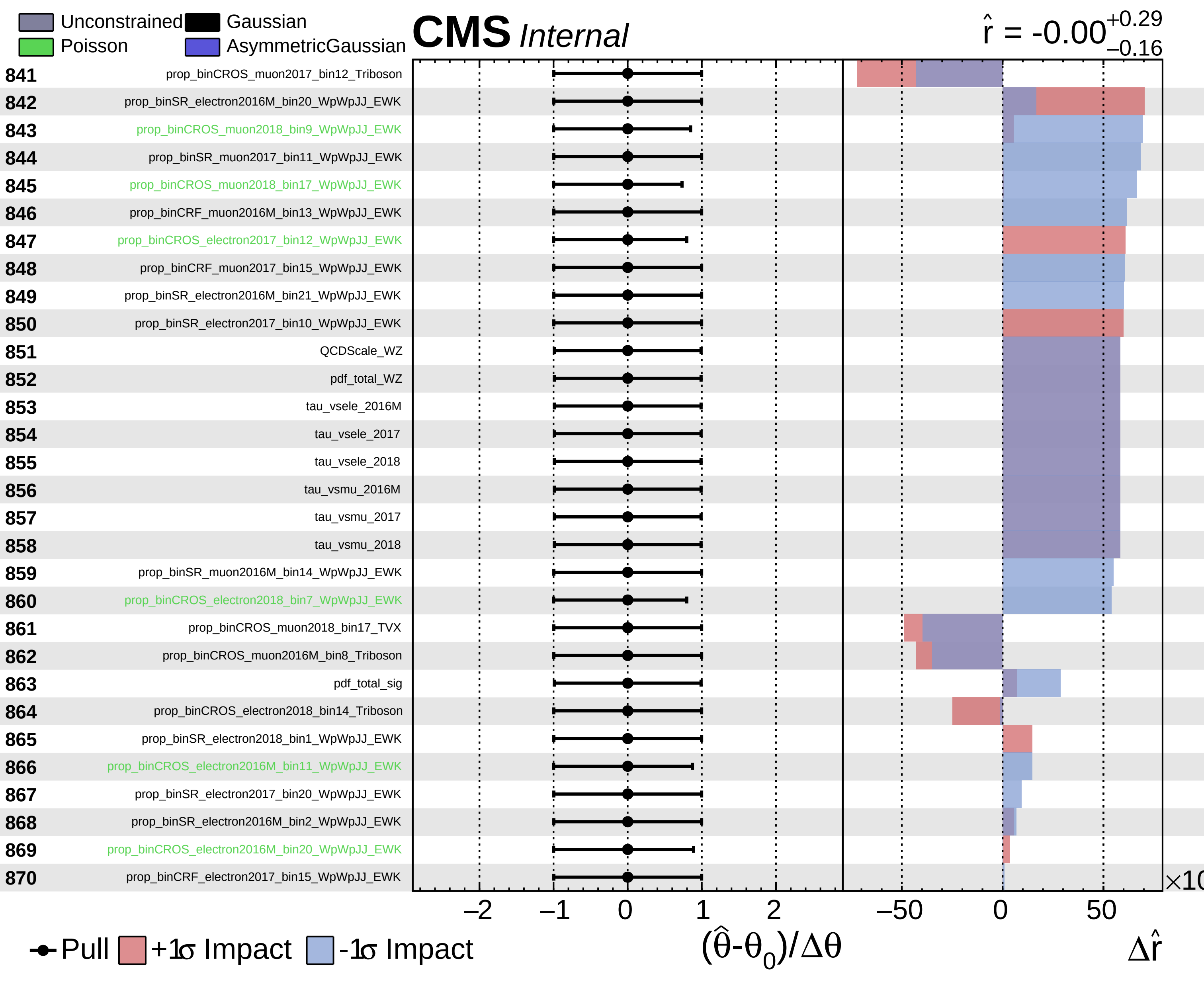


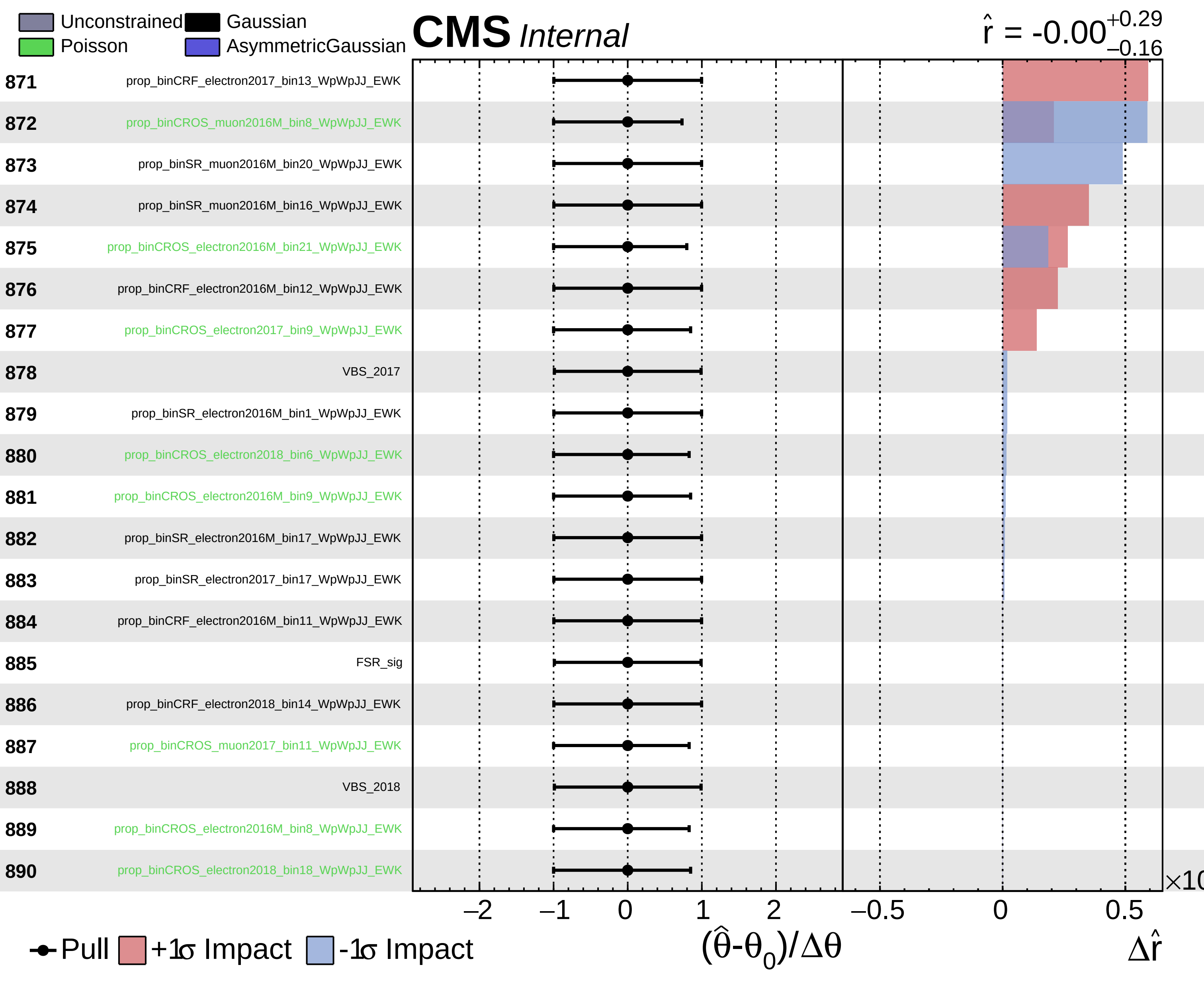
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.29}_{-0.16}$
751 prop_binCRF_electron2016M_bin11_TVX
752 prop_binSR_electron2016M_bin1_Triboson
753 prop_binSR_electron2016M_bin2_Triboson
754 prop_binSR_electron2018_bin1_WpWpJJ_QCD
755 prop_binSR_electron2017_bin20_WrongSign
756 prop_binCRF_electron2016M_bin10_ZZtoLep
757 prop_binCRF_electron2016M_bin11_Triboson
758 prop_binCRF_electron2016M_bin11_WpWpJJ_QCD
759 FES
760 prop_binCROS_muon2017_bin5_VG
761 prop_binCROS_electron2018_bin18_Triboson
762 prop_binCROS_muon2016M_bin9_Triboson
763 prop_binCROS_electron2018_bin18_ZZtoLep
764 prop_binCROS_electron2016M_bin21_Triboson
765 prop_binCROS_muon2017_bin11_TVX
766 prop_binCROS_electron2018_bin13_VG
767 prop_binCROS_electron2016M_bin20_Triboson
768 prop_binCROS_electron2016M_bin8_Triboson
769 prop_binCROS_electron2016M_bin11_TVX
770 prop_binCROS_electron2016M_bin11_Triboson
771 prop_binCROS_muon2017_bin11_ZZtoLep
772 prop_binCROS_electron2016M_bin11_ZZtoLep
773 prop_binCROS_electron2016M_bin20_TVX
774 prop_binCROS_muon2016M_bin9_VG
775 prop_binCROS_muon2018_bin11_TVX
776 prop_binCROS_electron2016M_bin21_TVX
777 prop_binCROS_electron2016M_bin21_ZZtoLep
778 prop_binCROS_muon2017_bin11_Triboson
779 prop_binCROS_muon2016M_bin9_TVX
780 prop_binCROS_electron2017_bin12_Triboson
−2 −1 0 1 2
−2 −1 0 1 2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.29}_{-0.16}$
781 prop_binCROS_muon2017_bin5_WZ
782 prop_binCROS_electron2017_bin10_ZZtoLep
783 prop_binCROS_electron2016M_bin20_ZZtoLep
784 prop_binCROS_muon2017_bin12_TVX
785 prop_binCROS_electron2018_bin13_Triboson
786 prop_binCROS_electron2018_bin13_TVX
787 prop_binCROS_electron2016M_bin8_ZZtoLep
788 prop_binCROS_electron2018_bin6_TVX
789 prop_binCROS_electron2018_bin13_ZZtoLep
790 prop_binCROS_muon2017_bin12_ZZtoLep
791 prop_binCROS_electron2017_bin10_Triboson
792 prop_binCROS_electron2017_bin12_ZZtoLep
793 prop_binCROS_muon2017_bin5_TVX
794 prop_binCROS_electron2018_bin9_ZZtoLep
795 prop_binCROS_electron2017_bin10_TVX
796 prop_binCROS_electron2016M_bin9_TVX
797 prop_binCROS_electron2018_bin6_Triboson
798 prop_binCROS_electron2016M_bin8_TVX
799 prop_binCROS_muon2016M_bin9_ZZtoLep
800 prop_binCROS_electron2017_bin9_ZZtoLep
801 prop_binCROS_muon2018_bin9_TVX
802 prop_binCROS_muon2017_bin5_ZZtoLep
803 prop_binCROS_electron2017_bin9_Triboson
804 prop_binCROS_electron2017_bin9_TVX
805 prop_binCROS_muon2018_bin9_ZZtoLep
806 prop_binCROS_electron2016M_bin9_ZZtoLep
807 prop_binCROS_electron2016M_bin9_Triboson
808 prop_binCROS_muon2017_bin5_Triboson
809 prop_binCROS_electron2018_bin9_TVX
810 prop_binCROS_muon2018_bin11_ZZtoLep
−2 −1 0 1 2
−0.2 0 0.2
×10
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.29}_{-0.16}$
811 prop_binCROS_electron2018_bin6_ZZtoLep
812 prop_binCROS_electron2017_bin12_TVX
813 prop_binCROS_muon2018_bin17_ZZtoLep
814 QCDScale_sig
815 prop_binCROS_electron2018_bin14_TVX
816 prop_binCROS_electron2018_bin14_ZZtoLep
817 prop_binCROS_muon2016M_bin8_TVX
818 prop_binCROS_electron2018_bin7_ZZtoLep
819 prop_binCROS_muon2016M_bin8_ZZtoLep
820 prop_binCROS_electron2018_bin7_TVX
821 prop_binCROS_muon2016M_bin8_VG
822 prop_binCROS_electron2018_bin14_WpWpJJ_EWK
823 prop_binCROS_muon2018_bin9_Triboson
824 prop_binCROS_electron2018_bin13_WpWpJJ_EWK
825 prop_binCROS_muon2018_bin17_Triboson
826 prop_binCROS_electron2018_bin9_Triboson
827 prop_binCROS_muon2018_bin11_Triboson
828 prop_binSR_electron2018_bin19_WpWpJJ_EWK
829 ISR_sig
830 VBS_2016M
831 prop_binCROS_electron2018_bin7_Triboson
832 prop_binCRF_electron2017_bin14_WpWpJJ_EWK
833 prop_binCRF_electron2016M_bin10_WpWpJJ_EWK
834 prop_binSR_muon2016M_bin17_WpWpJJ_EWK
835 prop_binCRF_electron2016M_bin17_WpWpJJ_EWK
836 prop_binSR_electron2018_bin21_WpWpJJ_EWK
837 prop_binCROS_muon2017_bin5_WpWpJJ_EWK
838 prop_binSR_muon2017_bin17_WpWpJJ_EWK
839 prop_binCRF_muon2017_bin12_WpWpJJ_EWK
840 prop_binCROS_muon2016M_bin9_WpWpJJ_EWK
−2 −1 0 1 2
−0.2 0 0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.29}_{-0.16}$
841 prop_binCROS_muon2017_bin12_Triboson
842 prop_binSR_electron2016M_bin20_WpWpJJ_EWK
843 prop_binCROS_muon2018_bin9_WpWpJJ_EWK
844 prop_binSR_muon2017_bin11_WpWpJJ_EWK
845 prop_binCROS_muon2018_bin17_WpWpJJ_EWK
846 prop_binCRF_muon2016M_bin13_WpWpJJ_EWK
847 prop_binCROS_electron2017_bin12_WpWpJJ_EWK
848 prop_binCRF_muon2017_bin15_WpWpJJ_EWK
849 prop_binSR_electron2016M_bin21_WpWpJJ_EWK
850 prop_binSR_electron2017_bin10_WpWpJJ_EWK
851 QCDScale_WZ
852 pdf_total_WZ
853 tau_vsele_2016M
854 tau_vsele_2017
855 tau_vsele_2018
856 tau_vsmu_2016M
857 tau_vsmu_2017
858 tau_vsmu_2018
859 prop_binSR_muon2016M_bin14_WpWpJJ_EWK
860 prop_binCROS_electron2018_bin7_WpWpJJ_EWK
861 prop_binCROS_muon2018_bin17_TVX
862 prop_binCROS_muon2016M_bin8_Triboson
863 pdf_total_sig
864 prop_binCROS_electron2018_bin14_Triboson
865 prop_binSR_electron2018_bin1_WpWpJJ_EWK
866 prop_binCROS_electron2016M_bin11_WpWpJJ_EWK
867 prop_binSR_electron2017_bin20_WpWpJJ_EWK
868 prop_binSR_electron2016M_bin2_WpWpJJ_EWK
869 prop_binCROS_electron2016M_bin20_WpWpJJ_EWK
870 prop_binCRF_electron2017_bin15_WpWpJJ_EWK
−2
−1
0
1
2
−50
0
50
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.29}_{-0.16}$
871 prop_binCRF_electron2017_bin13_WpWpJJ_EWK
872 prop_binCROS_muon2016M_bin8_WpWpJJ_EWK
873 prop_binSR_muon2016M_bin20_WpWpJJ_EWK
874 prop_binSR_muon2016M_bin16_WpWpJJ_EWK
875 prop_binCROS_electron2016M_bin21_WpWpJJ_EWK
876 prop_binCRF_electron2016M_bin12_WpWpJJ_EWK
877 prop_binCROS_electron2017_bin9_WpWpJJ_EWK
878 VBS_2017
879 prop_binSR_electron2016M_bin1_WpWpJJ_EWK
880 prop_binCROS_electron2018_bin6_WpWpJJ_EWK
881 prop_binCROS_electron2016M_bin9_WpWpJJ_EWK
882 prop_binSR_electron2016M_bin17_WpWpJJ_EWK
883 prop_binSR_electron2017_bin17_WpWpJJ_EWK
884 prop_binCRF_electron2016M_bin11_WpWpJJ_EWK
885 FSR_sig
886 prop_binCRF_electron2018_bin14_WpWpJJ_EWK
887 prop_binCROS_muon2017_bin11_WpWpJJ_EWK
888 VBS_2018
889 prop_binCROS_electron2016M_bin8_WpWpJJ_EWK
890 prop_binCROS_electron2018_bin18_WpWpJJ_EWK
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$